

Male-Biased Cyp17a2 Governs Antiviral Sexual Dimorphism in Fish via STING Stabilization and Viral Protein Degradation

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This **valuable** study describes an interesting infection phenotype that differs between adult male and female zebrafish. The authors present data indicating that male-biased expression of Cyp17a2 mediates viral infection through STING and USP8 activity regulation. The authors present **solid** evidence linking this factor to direct and indirect antiviral outcomes through ubiquitination pathways. These findings raise interesting questions about immune mechanisms that underlie sex dimorphism and the selective pressures that might shape it.

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Abstract

Differences in immunity between males and females in living organisms are generally thought to be due to sex hormones and sex chromosomes, and it is often assumed that males have a weaker immune response. Here we report that in fish, males exhibit stronger antiviral immune responses, which are driven by the male-biased gene Cyp17a2 rather than by hormones or sex chromosomes. First, we observed that male zebrafish exhibit enhanced antiviral resistance compared to females, and notably, zebrafish lack sex chromosomes. Through transcriptomic screening, we found that *cyp17a2* was specifically highly expressed in male fish. *Cyp17a2* knockout males were equivalent to wild-type males in terms of sex organs and androgen secretion, but the ability to upregulate IFN as well as antiviral resistance was greatly reduced. Then, Cyp17a2 is identified as a positive IFN regulator which located at endoplasmic reticulum, and specifically interacts with and enhances STING mediated antiviral responses. Mechanistically, Cyp17a2 stabilizes STING expression by

recruiting the E3 ubiquitin ligase bloodthirsty-related gene family member 32 (btr32), which facilitates K33-linked polyubiquitination. The capacity of IFN induction of Cyp17a2 was abolished when STING is knockdown. Meanwhile, Cyp17a2 also attenuates viral infection directly to strengthen the antiviral capacity as an antiviral protein, Cyp17a2 degrades the spring viremia of carp virus (SVCV) P protein by utilizing USP8 to reduce its K33-linked polyubiquitination. These findings reveal a sex-based regulatory mechanism in teleost antiviral immunity, broadening our understanding of sexual dimorphism in immune responses beyond the conventional roles of sex chromosomes and hormones.

Introduction

Research has demonstrated that innate immune responses to viruses differ significantly between the sexes. Males exhibit heightened susceptibility to viral pathogens, correlating with their generally weaker innate immune activation compared to females¹. This immunological dimorphism manifests clinically as elevated infection intensity (quantified by intra-host viral load) and prevalence (population-level infection rates) in males². Notably, sex-specific clinical outcomes are exemplified by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infections, where epidemiological data suggest males face twice the risk of intensive care unit (ICU) admission and a 30% higher mortality rate compared to females³. Similar trends extend to other viral pathogens including Dengue virus, hantaviruses, and hepatitis B/C viruses, all exhibiting male-biased infection prevalence^{4, 5, 6}.

Several factors have been identified as contributors to these sex-specific disparities in antiviral immunity, including hormonal variations and sex chromosomes. For instance, androgens such as testosterone exhibit broad immunosuppressive effects, including reduced NK cell cytotoxicity, impaired neutrophil chemotaxis, and attenuated macrophage proinflammatory cytokine production *in vitro*⁷. In addition, the Y chromosome harbors only two annotated miRNAs, while the X chromosome contains numerous immune-relevant miRNAs, and emerging evidence implicates X-escapee miRNAs as key modulators of sex-biased immune responses^{8, 9}.

It is well known that the interferon (IFN) response is an important host immune response to viral infection, in both mammals and fish¹⁰. The signaling pathways that activate IFN expression, although slightly different in fish and mammals, are generally conserved in both¹¹. The endoplasmic reticulum (ER) protein stimulator of IFN genes (STING) is an important antiviral protein capable of resisting both DNA and RNA viruses by activating TANK binding kinase 1 (TBK1) during the induction of IFN production, which in turn phosphorylates IFN regulatory Factor 3 (IRF3) in the nucleus and activates the signaling of IFN transcription^{12, 13}. For instance, STING deficiency has been shown to impair innate immune responses against multiple RNA viruses, such as dengue virus, West Nile virus, and Japanese encephalitis virus^{14, 15, 16}. Notably, oncogenes of the DNA tumor viruses, including human papillomavirus E7 and adenovirus E1A, function as potent and specific inhibitors of the cyclic GMP-AMP synthase (cGAS)-STING signaling pathway¹⁷.

Cyp17a is classified within the cytochrome P450 (CYP450) family. In mammals, there is only one form, cyp17a1, which possesses both hydroxylase and lyase functions. Mutations in the human *cyp17a1* gene are associated with congenital adrenal hyperplasia (CAH)¹⁸. Comparative genomic analyses reveal that teleost fishes possess two distinct *cyp17a* isoforms, the evolutionarily conserved *cyp17a1* and a phylogenetically divergent *cyp17a2* paralog, with the latter representing a teleost-specific innovation absent in mammalian genomes¹⁹. Current investigations of Cyp17a2 in teleost species have predominantly centered on its mechanistic roles governing sexual development and differentiation, for example, with knockout of zebrafish *cyp17a2* leads to significant impairment of sperm motility²⁰. Although Cyp17a2 has been characterized in sexual development, its non-canonical physiological roles have yet to be fully elucidated.

In most vertebrates, the XY sex chromosomes exhibit significant physiological differences between males and females, as exemplified in humans. However, the sex determination mechanisms in fish are more complex. For instance, zebrafish lack sex chromosomes, which confers a natural advantage. This absence allows for the exclusion of sex chromosome-related differences in the study of immunological disparities between males and females. Consequently, we selected zebrafish as the model organism for this investigation. This enabled us to more precisely identify the genes that are differentially expressed in male and female autosomes, which is crucial for understanding the immunological differences between the sexes. This study elucidates the role of the teleost-specific gene *cyp17a2* in antiviral immunity. Furthermore, the discovery that the male-biased expression of *cyp17a2* confers enhanced resistance to infection in males provides valuable insight into the complex mechanisms underlying sexual immune dimorphism.

Results

1. Male fish exhibit enhanced resistance to viral infection compared to female fish

To determine whether sex differences exist in the antiviral response capacity of fish, age-matched female and male zebrafish were intraperitoneally (i.p.) injected with SVCV. Male fish exhibited a significantly higher survival rate than female fish within seven days post-infection (**Fig. 1A**). Concurrently, females demonstrated a markedly higher incidence of cutaneous hemorrhages relative to males (**Fig. 1B**). Consistent with the reduced mortality, male fish showed less tissue damage in the liver and spleen following viral infection (**Fig. 1C**). IF analysis revealed that the SVCV N protein, represented by green fluorescence, was barely detectable in male tissues but was prominently present in female tissues (**Fig. 1D**). At the mRNA level, viral transcripts were quantified in these tissues, and the abundance of *svc-v-n* gene was significantly lower in male tissues (**Fig. 1E**). At the protein level, SVCV N and G proteins were less frequently detected in male tissues compared to female tissues in 7 samples (**Fig. 1F**). Additionally, viral titers were significantly lower in the hearts of male fish compared to female fish (**Fig. 1G**). These findings demonstrate a stronger antiviral response capacity in male fish compared to female fish.

2. Male-biased gene *cyp17a2* orchestrates antiviral immunity through RLR pathway

To explore the potential causes of the observed differences in virus resistance between male and female fish, transcriptome sequencing was performed on head-kidney tissues of healthy hermaphroditic adult zebrafish. Differential expression analysis identified 1511 upregulated genes and 1117 downregulated genes (**Fig. 2A** and Table S2). Among these, *cyp17a2* exhibits significant male-biased upregulation in eight sex-related genes, and it was confirmed by subsequent qPCR (**Fig. 2B** and S1A). Furthermore, we systematically quantified Cyp17a2 mRNA and protein expression in multiple tissues of male and female zebrafish, demonstrating a consistent male-biased expression pattern of Cyp17a2 (**Fig. 2C** and 2D). Thus, Cyp17a2 was selected for further study. We first sought to understand whether it was associated with host antiviral immunity, and the significant upregulation of the Cyp17a2 at the mRNA and protein level in response to viral stimulation suggested its possible involvement in the antiviral process (**Fig. S1B** and **S1C**). Following generation of *cyp17a2* knockout zebrafish, previous study revealed that male *cyp17a2*^{-/-} mutants displayed normal development of male-typical secondary sexual characteristics comparable to WT males. However, female *cyp17a2*^{-/-} mutants exhibited significantly underdeveloped genital papillae compared to their WT counterparts. To eliminate potential confounding effects of androgen-mediated pathways on immune parameters, subsequent

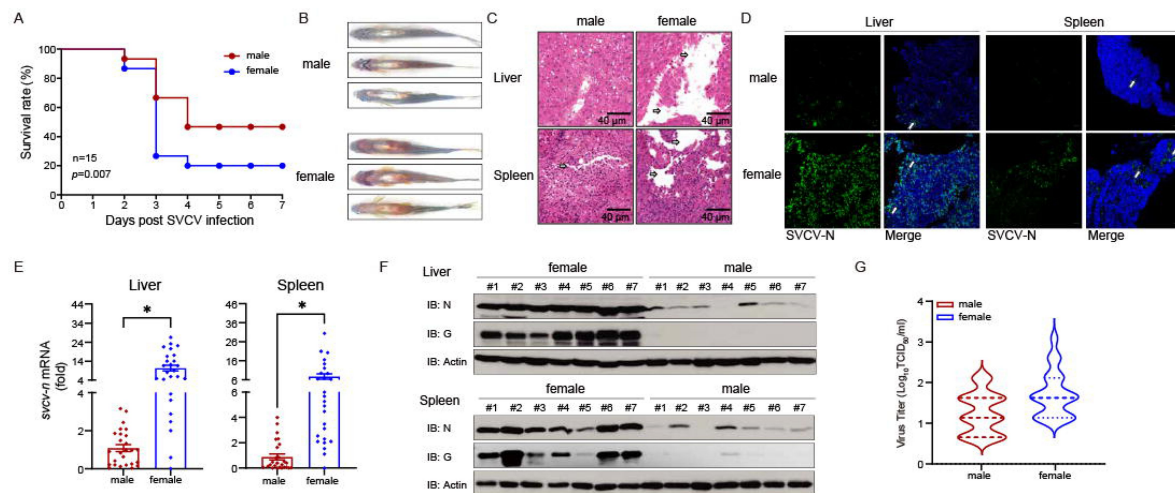


Figure 1.

Male fish show stronger viral resistance than females.

(A) Survival (Kaplan-Meier Curve) of male and female zebrafish ($n = 15$ per group) at various days after i.p. injected with SVCV (5×10^8 TCID₅₀/ml, 5 μ L/individual). (B) Sex-specific morphological alterations in zebrafish given i.p. injection of SVCV for 48 h ($n = 3$ per group). (C) Microscopy of H&E-stained liver and spleen sections from male and female zebrafish treated with SVCV for 48 h. (D) IF analysis of SVCV-N protein in liver and spleen of male and female zebrafish treated with SVCV for 48 h. (E) qPCR analysis of *svc-v* mRNA in the liver and spleen of male and female zebrafish ($n = 26$ per group) given i.p. injection of SVCV for 48 h. (F) IB analysis of SVCV proteins in the liver and spleen sections of male and female zebrafish ($n = 7$ per group) treated with SVCV for 48 h. (G) The viral titer of heart in male and female zebrafish ($n = 40$ per group) treated with SVCV for 48 h.

experiments were conducted exclusively using male specimens from both genotypes. The male *cyp17a2*^{-/-} mutants exhibited a higher mortality rate than WT males after viral infection (**Fig. 2E** [↗](#)).

Significant tissue damage and strong green fluorescence representing the SVCV N protein was observed in the *cyp17a2*^{-/-} tissues, virus titers assay also confirmed that viral proliferation was higher in *cyp17a2*^{-/-} group (Fig. S1D-S1F). Consistent with these observations, the replication of viral genes transcription and protein levels were increased in *cyp17a2*^{-/-} homozygote tissues, these data demonstrated that Cyp17a2 is crucial for host antiviral defense (**Fig. 2F** [↗](#) and S1G). To further understand their functional mechanisms, a transcriptomic analysis contains total of 3,114 differentially expressed genes (DEGs) were identified, there are 1,217 genes upregulated and 1,897 were downregulated in the liver of *cyp17a2* knockout male zebrafish compared to WT males after viral infection (Fig. S1H). Kyoto encyclopedia of genes and genomes (KEGG) pathway analysis revealed RIG-I-like receptors (RLRs), NOD-like receptors (NLRs), and Toll-like receptors (TLRs) signaling pathways were significantly affected (Fig. S1I). Gene set enrichment analysis (GSEA) showed that RLRs were downregulated in *cyp17a2*^{-/-} group under SVCV infection, combining with numerous IFN-stimulated genes (ISGs) decreases (Fig. S1J and S2). To validate the transcriptome data, *ifn ϕ 1* levels were assessed and identified significantly downregulated in the *cyp17a2*^{-/-} group (**Fig. 2G** [↗](#)). Collectively, these data suggest that viral proliferation was enhanced and IFN response were reduced in *cyp17a2*^{-/-} fish.

3. Cyp17a2 enhances IFN expression and suppresses viral proliferation

Given the critical role of the IFN response in host defense against viral infections, we investigated the impact of Cyp17a2 on IFN expression and viral infection *in vitro*. Overexpression of Cyp17a2 significantly upregulated the IFN promoter and ISRE activities in response to poly I:C or SVCV stimulation (**Fig. 3A** [↗](#) and S3A-S3B). An effective sh-*cyp17a2*#2 was generated and confirmed at both the mRNA and protein levels, and knockdown of *cyp17a2* inhibited both the IFN promoter and ISRE activity (**Fig. 3B-3E** [↗](#) and S3C-S3D). RNA sequencing analysis of the cell line revealed that SVCV infection activated numerous ISGs in Cyp17a2-overexpressing cells, whereas these ISGs were significantly suppressed upon *cyp17a2* knockdown (**Fig. 3F** [↗](#) and Table S3). The induction or attenuation of IFN and ISGs upon Cyp17a2 overexpression or knockdown were confirmed by qPCR analysis, respectively (**Fig. 3G** [↗](#) and S3E). For antiviral capacity assays, Cyp17a2 from zebrafish and gibel carp both inhibited the proliferation of host-specific viruses, conversely, *cyp17a2* knockdown significantly enhanced viral proliferation (**Fig. 3H** [↗](#) and S3F). Regarding viral mRNA and proteins, overexpression or knockdown Cyp17a2 also displayed the similar opposite findings (**Fig. 3I-3J** [↗](#) and S3G-S3H). IF assays indicated a lower intensity of green signals representing the viral protein in the Cyp17a2 overexpression group compared to the control, while a higher green signal was observed in the *cyp17a2* knockdown group relative to the normal group (**Fig. 3K** [↗](#) and S3I). These results suggest that Cyp17a2 upregulates IFN expression and enhances antiviral capacity in the host.

4. Cyp17a2 interacts with STING and stabilizes its expression

Given the critical role of the RLR signaling pathway in fish IFN activation, the relationship between Cyp17a2 and RLR signaling adaptors was investigated. Co-transfection with Myc-Cyp17a2 and Flag-tagged RLR components (MAVS, TBK1, STING, IRF3, IRF7) followed by anti-Flag immunoprecipitation identified specific Cyp17a2-STING interaction via anti-Myc detection (**Fig. 4A** [↗](#)). Reciprocal Co-IP and semi-endogenous IP assays confirmed this interaction (**Fig. 4B** [↗](#) and 4C). Truncated STING mutants were generated to map binding domains (**Fig. 4D** [↗](#)). The transmembrane domain (TM)-deficient STING-ΔTM-HA failed to bind Cyp17a2, indicating TM domain dependence (**Fig. 4E** [↗](#)). Cyp17a2-GFP overlapped with ER markers in confocal microscopy and colocalized with STING but failed to colocalize with the ΔTM mutant (**Fig. 4F-4G** [↗](#) and S4A).

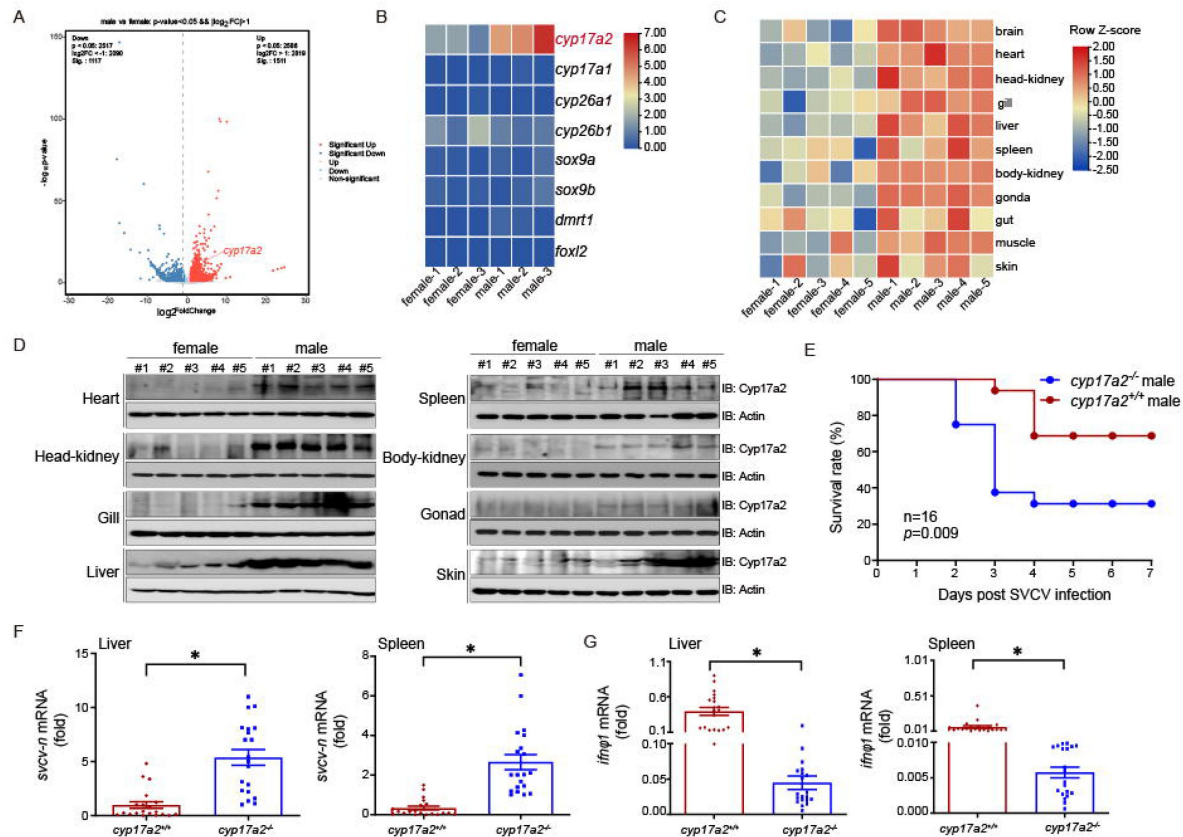


Figure 2.

Male-biased *cyp17a2* regulates RLR-mediated antiviral immunity.

(A) The differentially expressed gene number of mRNA variations presented as a volcano plot in head-kidney of male and female zebrafish. (B) Heatmap view of mRNA variations of eight sex-regulated genes in the head-kidney of male and female zebrafish. Values are represented as $\log_2(FPKM+1)$. (C) Heatmap view of *cyp17a2* mRNA in the brain, heart, head-kidney, gill, liver, spleen, body-kidney, gonad, gut, muscle, and skin of male and female zebrafish ($n = 5$ per group). Expression values [$\log_2(\text{fold change})$] were transformed to Z-scores across rows for each gene to highlight patterns of relative expression. (D) IB analysis of Cyp17a2 in the heart, head-kidney, gill, liver, spleen, body-kidney, gonad, and skin of male and female zebrafish ($n = 5$ per group). (E) Survival (Kaplan-Meier Curve) of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish ($n = 16$ per group) at various days after i.p. injected with SVCV (5×10^8 TCID₅₀/ml, 5 μ l/individual). (F) qPCR analysis of *svc-v-n* mRNA in the liver and spleen of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish ($n = 20$ per group) given i.p. injection of SVCV for 48 h. (G) qPCR analysis of *ifnφ1* mRNA in the liver and spleen of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish ($n = 20$ per group) given i.p. injection of SVCV for 48 h.

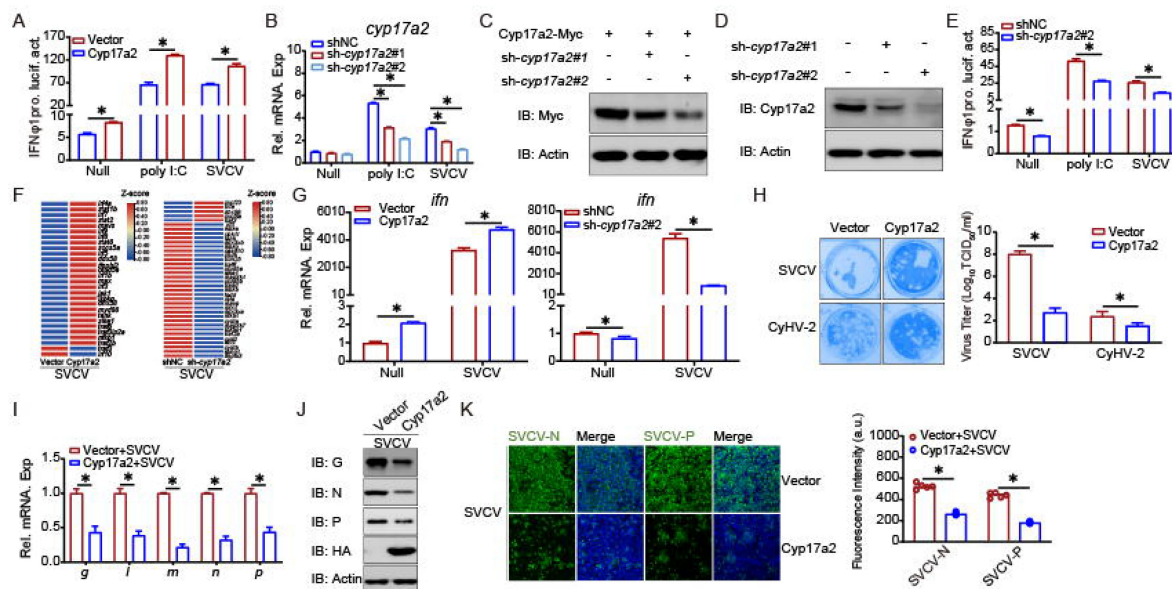


Figure 3.

Cyp17a2 upregulates IFN expression and inhibits viral replication.

(A and E) Luciferase activity of IFN̑1pro in EPC cells transfected with indicated plasmids for 24 h, and then untreated or transfected with poly I:C (0.5 µg) or infected with SVCV (MOI = 1) for 24 h before luciferase assays. (B) qPCR analysis of *cyp17a2* in EPC cells transfected with indicated plasmids for 24 h, and then untreated or transfected with poly I:C (2 µg) or infected with SVCV (MOI = 1) for 24 h. (C and D) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (F) Heatmap view of mRNA variations of SVCV-activated ISG sets in the Cyp17a2-overexpressing cells or *cyp17a2* knockdown cells and infected with SVCV for 24 h. Expression values are $\log_2(\text{FPKM}+1)$ transformed and presented as row Z-scores. (G) qPCR analysis of *ifn̑1* in EPC cells transfected with indicated plasmids for 24 h, and then untreated or infected with SVCV (MOI = 1) for 24 h. (H) Plaque assay of virus titers in EPC cells or GICB cells transfected with indicated plasmids for 24 h, followed by SVCV or CyHV-2 challenge for 24 h or 48 h. (I and J) qPCR and IB analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. (K) IF analysis of SVCV proteins in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h.

To identify the effect of Cyp17a2 on STING regulation, the mRNA level of *sting* was first analyzed, minimal change was observed in *sting* upon Cyp17a2 overexpression (Fig. 4H). Protein-level studies demonstrated Cyp17a2 remarkably enhanced STING abundance and confirmed by confocal microscopy (Fig. 4I and S4B). Endogenous STING was similarly stabilized dependent on Cyp17a2 under untreated or stimulated conditions (Fig. 4J and S4C). Cycloheximide (CHX) chase assay showed prolonged exogenous and endogenous STING half-life with Cyp17a2 co-expression, and it was also confirmed by knockdown experiments (Fig. 4K and S4D-S4E). These data demonstrate that Cyp17a2 binds STING via its TM and enhances STING posttranslational stability.

5. Cyp17a2 enhances STING-mediated IFN expression and antiviral responses

Considering that Cyp17a2 targets STING, a positive regulator of IFN production, we investigated whether Cyp17a2 influences the IFN expression induced by STING. The IFN ϕ 1pro activity induced by STING overexpression was significantly enhanced by Cyp17a2, with similar amplification observed for ISRE promoter activation (Fig. 5A and S5A). Conversely, *cyp17a2* knockdown was shown to impair STING-mediated IFN ϕ 1pro and ISRE activation (Fig. 5B and S5B). Corresponding increases in *ifn* and *vig1* transcripts were detected in Cyp17a2 overexpressing cells, while Cyp17a2 knockdown suppressed these responses (Fig. 5C-5D and S5C-S5D). The subsequent investigation focused on the potential modulatory effect of Cyp17a2 on the STING-induced cellular antiviral response. STING overexpression reduced cytopathogenic effect (CPE), with viral titers decreased 336-fold compared to the control group. This protection was augmented by Cyp17a2 co-expression, achieving an additional 20-fold reduction (Fig. 5E and S5E). IB and IF analysis confirmed that Cyp17a2 potentiated STING-mediated suppression of viral protein expression (Fig. 5F and S5F). SVCV gene transcripts were suppressed by STING, with enhanced inhibition upon Cyp17a2 co-expression (Fig. 5G). STING dependency was validated using shRNA knockdown (Fig. S5G). The antiviral capacity enhanced by Cyp17a2 was markedly attenuated in cells with *sting* knockdown, as evidenced by restored viral titers, gene and protein expression (Fig. 5H-5J and S5H-S5I). These results demonstrate that Cyp17a2 potentiates STING activity to enhance IFN production and antiviral responses.

6. Cyp17a2 stabilizes STING through btr32-mediated K33-linked polyubiquitination

To elucidate the molecular mechanism by which Cyp17a2 augments STING stabilization, proteasomal and lysosomal degradation were pharmacologically inhibited using MG132 and Chloroquine (CQ), respectively. Accelerated STING degradation caused by *cyp17a2* knockdown was rescued by MG132 but not CQ, suggesting knockdown of *cyp17a2* promoted STING degradation via the proteasome pathway (Fig. 6A). Then, enhanced polyubiquitination of STING was detected upon Cyp17a2 co-expression (Fig. 6B). Ubiquitin linkage analysis revealed selective enhancement of K33-linked chains, with *cyp17a2* knockdown shown to reduce both WT and K33-specific ubiquitination (Fig. 6C and S6A). Subsequently, candidate E3 ligases were screened, revealing btr32 and SOCS3a as STING interactors (Fig. 6D). btr32 was specifically identified as a Cyp17a2 binding partner through reciprocal Co-IP (Fig. 6E-6G). Then the subcellular localization of btr32 was monitored, btr32-GFP signal overlapped with the red signals of the ER marker, indicating that btr32 colocalizes with the ER (Fig. 6H). Since the results identify that both STING and Cyp17a2 are localized to the ER, btr32 was examined for cellular localization with both STING and Cyp17a2, and the anticipated colocalization pattern was confirmed (Fig. 6I and 6J). Cyp17a2 was found to strengthen btr32-STING interaction (Fig. 6K and S6B). Functional assays demonstrated that btr32 augmented of Cyp17a2-mediated IFN ϕ 1pro activity and *vig1* mRNA induction (Fig. 6L and 6M). btr32 potentiated the Cyp17a2-dependent stabilization of STING in both endogenous and exogenous systems (Fig. 6N). Two shRNAs target btr32 were designed and generated, and after validating the knockdown efficiencies for both endogenous and exogenous

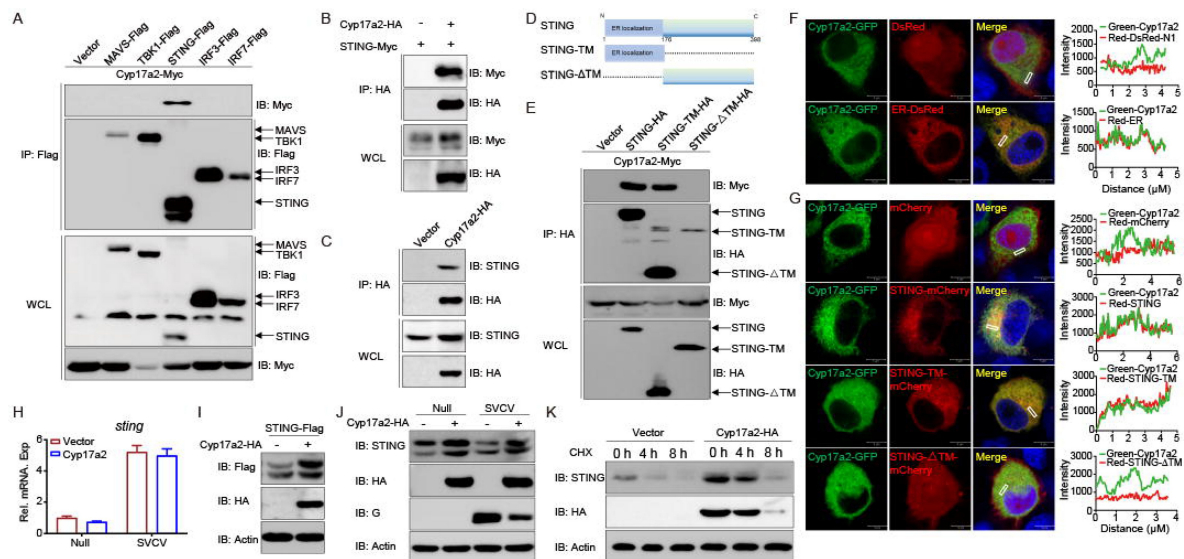


Figure 4.

Cyp17a2 physically interacts with STING and stabilizes its expression.

(A-C and E) IB analysis of WCLs and proteins immunoprecipitated with anti-Flag or HA Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (D) Schematic representation of full-length STING and its mutants. (F and G) Confocal microscopy of Cyp17a2 and ER or STING and its mutants in EPC cells transfected with indicated plasmids for 24 h. The coefficient of colocalization was determined by qualitative analysis of the fluorescence intensity of the selected area in Merge. (H) qPCR analysis of *sting* in EPC cells transfected with indicated plasmids for 24 h, followed by untreated or infected with SVCV (MOI = 1) for 24 h. (I) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (J) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h, followed by untreated or infected with SVCV (MOI = 1) for 24 h. (K) IB analysis of proteins in EPC cells transfected with indicated plasmids for 18 h, then treated with CHX for 4 h and 8 h.

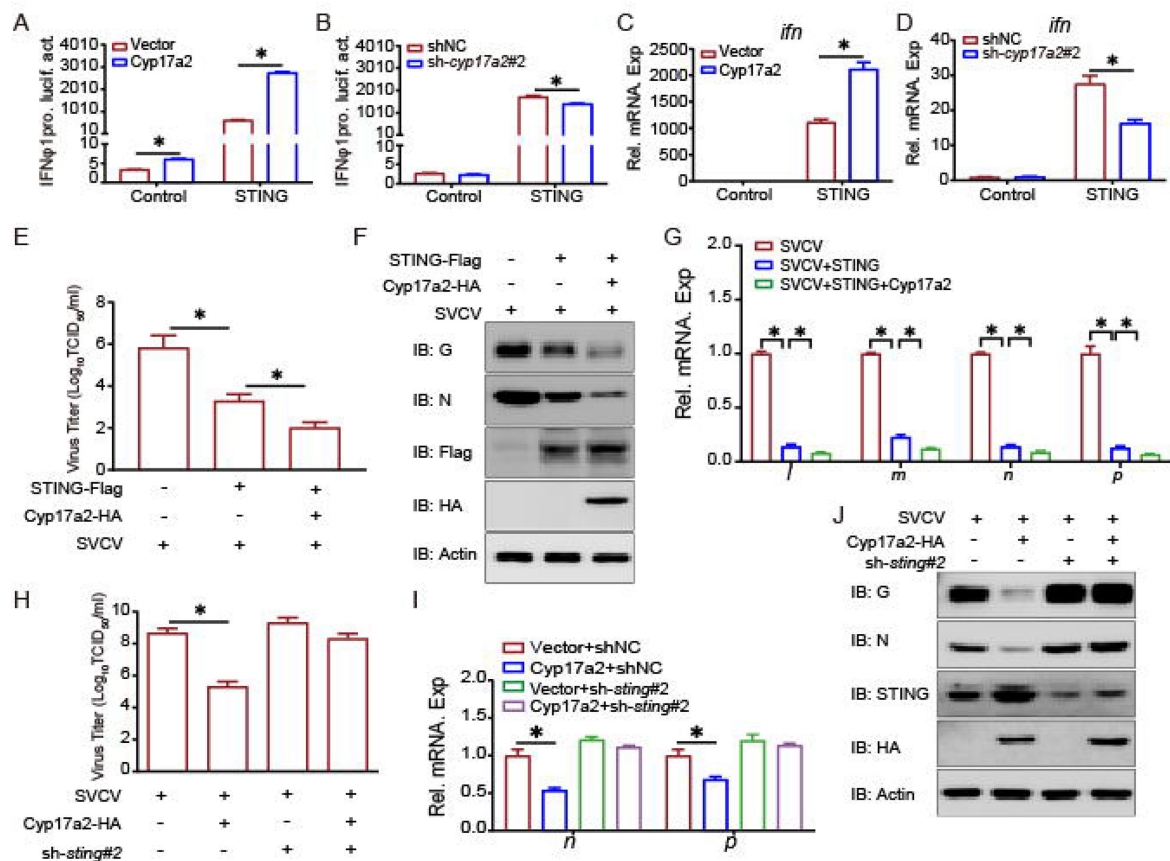


Figure 5.

Cyp17a2 potentiates STING-induced IFN production and antiviral responses.

(A and B) Luciferase activity of IFN ϕ 1pro in EPC cells transfected with indicated plasmids for 24 h before luciferase assays. (C and D) qPCR analysis of *ifn* in EPC cells transfected with indicated plasmids for 24 h and then infected with SVCV (MOI = 1) for 24 h. (E and H) Detection of virus titers in EPC cells transfected with indicated plasmids for 24 h and then infected with SVCV (MOI = 1) for 24 h. (F and J) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h and then untreated or infected with SVCV (MOI = 1) for 24 h. (G and I) qPCR analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h.

targets, *sh-btr32#2* was selected for the subsequent assay (Fig. S6C–S6D). The knockdown of *btr32* impaired Cyp17a2-enhanced STING activity, as shown by reduced IFN β 1pro response and *vig1* transcripts (Fig. S6E and S6F). Additionally, *btr32* knockdown abolished Cyp17a2-mediated STING upregulation in endogenous and overexpression system (Fig. S6G). These findings establish that Cyp17a2 recruits *btr32* to catalyze K33-linked polyubiquitination, thereby stabilizing STING protein.

7. STING K203 is essential for Cyp17a2-mediated ubiquitination

The mechanism between *btr32* and Cyp17a2 in STING stabilization was further characterized. *btr32* overexpression was shown to intensify Cyp17a2-driven STING ubiquitination, while *btr32* knockdown attenuated this effect (Fig. 7A and S7A). Specific enhancement of K33-linked ubiquitination by *btr32* was demonstrated, with reciprocal reduction upon *btr32* abrogation (Fig. 7B and S7B). Mass spectrometry analysis identified K203 and K221 as candidate ubiquitination sites (Fig. 7C). Among generated lysine-to-arginine mutants (K203R, K221R), STING stabilization by Cyp17a2 was abolished specifically in K203R mutant (Fig. 7D). Corresponding loss of IFN induction and ubiquitination capacity were observed in K203R mutant (Fig. 7E–7G). K33-linked ubiquitination was specifically impaired in K203R but not K221R variants (Fig. S7C). *btr32*-mediated enhancement of WT and K33-specific ubiquitination was eliminated in K203R mutant (Fig. 7H and S7D). *btr32* domain analysis was performed using RING/BBOX/SPRY deletion mutants (Fig. 7I). The SPRY domain was required for *btr32*-STING-Cyp17a2 interactions (Fig. 7J–7L). Both RING and SPRY domains were essential for *btr32*-driven ubiquitination (Fig. 7M and S7E). These data establish K203 as the critical ubiquitination site in STING and define RING/SPRY domains of *btr32* as necessary for Cyp17a2-mediated stabilization.

8. Cyp17a2 Degrades SVCV P Protein Through Modulation of K33-Linked Polyubiquitination

To determine whether Cyp17a2 directly engage in antiviral activity beyond regulating host immune responses, Co-IP assays were performed. Transient transfection experiments revealed specific interaction between Cyp17a2 and SVCV P protein, which was confirmed reciprocally (Fig. 8A and 8B). This interaction was further observed in SVCV-infected cells using P protein-specific antibody (Fig. 8C). In our antiviral trials, overexpression of Cyp17a2 led to a reduction in P protein levels during SVCV infection. Consequently, the investigation focused on whether Cyp17a2 could directly target the SVCV P protein for degradation. As anticipated, the co-expression of Cyp17a2 resulted in a decrease in the level of SVCV P protein. Overexpression of Cyp17a2 reduced P protein levels during SVCV infection (Fig. 8D). Fluorescence microscopy confirmed diminished red fluorescent signals corresponding to SVCV P protein in Cyp17a2-expressing cells (Fig. 8E). Confocal microscopy analysis of SVCV P protein subcellular distribution revealed that the fluorescent signals from P-mCherry fusion constructs (red) showed partial co-localization with Cyp17a2 (green) in cytoplasmic compartments (Fig. 8F). To investigate degradation mechanisms, inhibitors targeting distinct pathways were tested. Cyp17a2-mediated P protein degradation was blocked by the proteasome inhibitor MG132, but not by autophagy inhibitors (3-MA, Baf-A1, CQ) (Fig. 8G). Ubiquitination assays demonstrated that Cyp17a2 significantly reduced P protein polyubiquitination (Fig. 8H). Mutation analysis using ubiquitin variants revealed that Cyp17a2 selectively decreased K33-linked polyubiquitination, as K33R mutants exhibited no significant changes, while K33-specific ubiquitination was markedly reduced (Fig. 8I and 8J). In contrast, knockdown of *cyp17a2* exhibited minimal impact on both WT-Ub and K33-linked polyubiquitination levels of the P protein (Fig. 8K). These findings demonstrate that Cyp17a2 promotes proteasomal degradation of the SVCV P protein by selectively attenuating K33-linked polyubiquitination, defining its role as an antiviral protein.

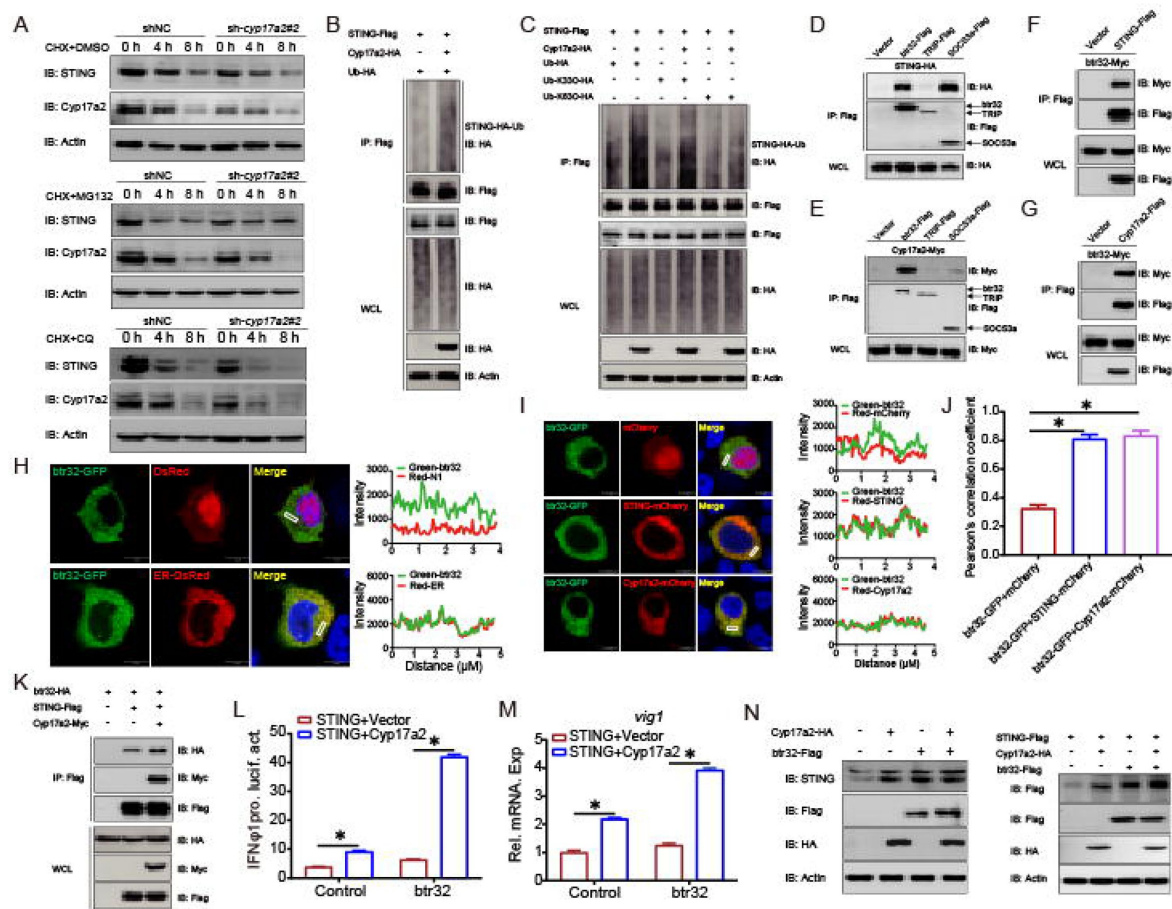


Figure 6.

Cyp17a2 stabilizes STING by orchestrating btr32-catalyzed K33-linked polyubiquitination.

(A) IB analysis of proteins in EPC cells transfected with indicated plasmids for 18 h, then treated with CHX and DMSO, MG132, or CQ for 4 h and 8 h. (B and C) STING ubiquitination assays in EPC cells transfected with indicated plasmids for 24 h. (D-G, K) IB analysis of WCLs and proteins immunoprecipitated with anti-Flag Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (H and I) Confocal microscopy of btr32 and ER or STING or Cyp17a2 in EPC cells transfected with indicated plasmids for 24 h. The coefficient of colocalization was determined by qualitative analysis of the fluorescence intensity of the selected area in Merge. (J) Colocalization analyses of [figure 6I](#) were performed by calculating Pearson correlation coefficient. (L and M) Luciferase activity of IFN ϕ 1pro and qPCR analysis of *vig1* in EPC cells transfected with indicated plasmids for 24 h. (N) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h.

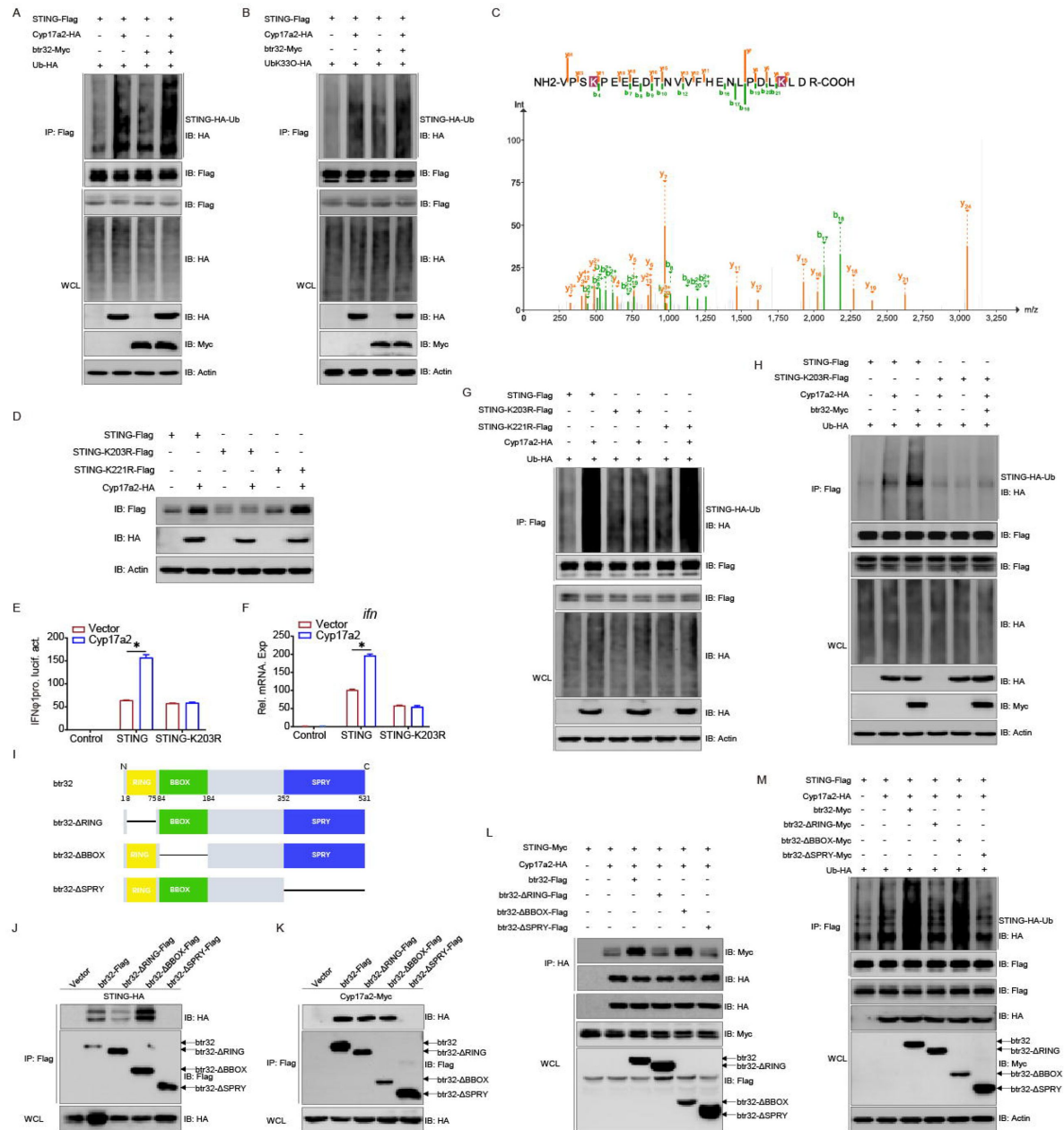


Figure 7

Cyp17a2-mediated ubiquitination of STING is dependent on the K203.

(A and B, G and H, and M) STING ubiquitination assays in EPC cells transfected with indicated plasmids for 24 h. (C) Mass spectrometry analysis of a peptide derived from ubiquitinated STING-Myc. (D) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (E and F) Luciferase activity of IFN ϕ 1pro and qPCR analysis of *ifn* in EPC cells transfected with indicated plasmids for 24 h. (I) Schematic representation of full-length btr32 and its mutants. (J-L) IB of WCLs and proteins immunoprecipitated with anti-Flag/HA Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h.

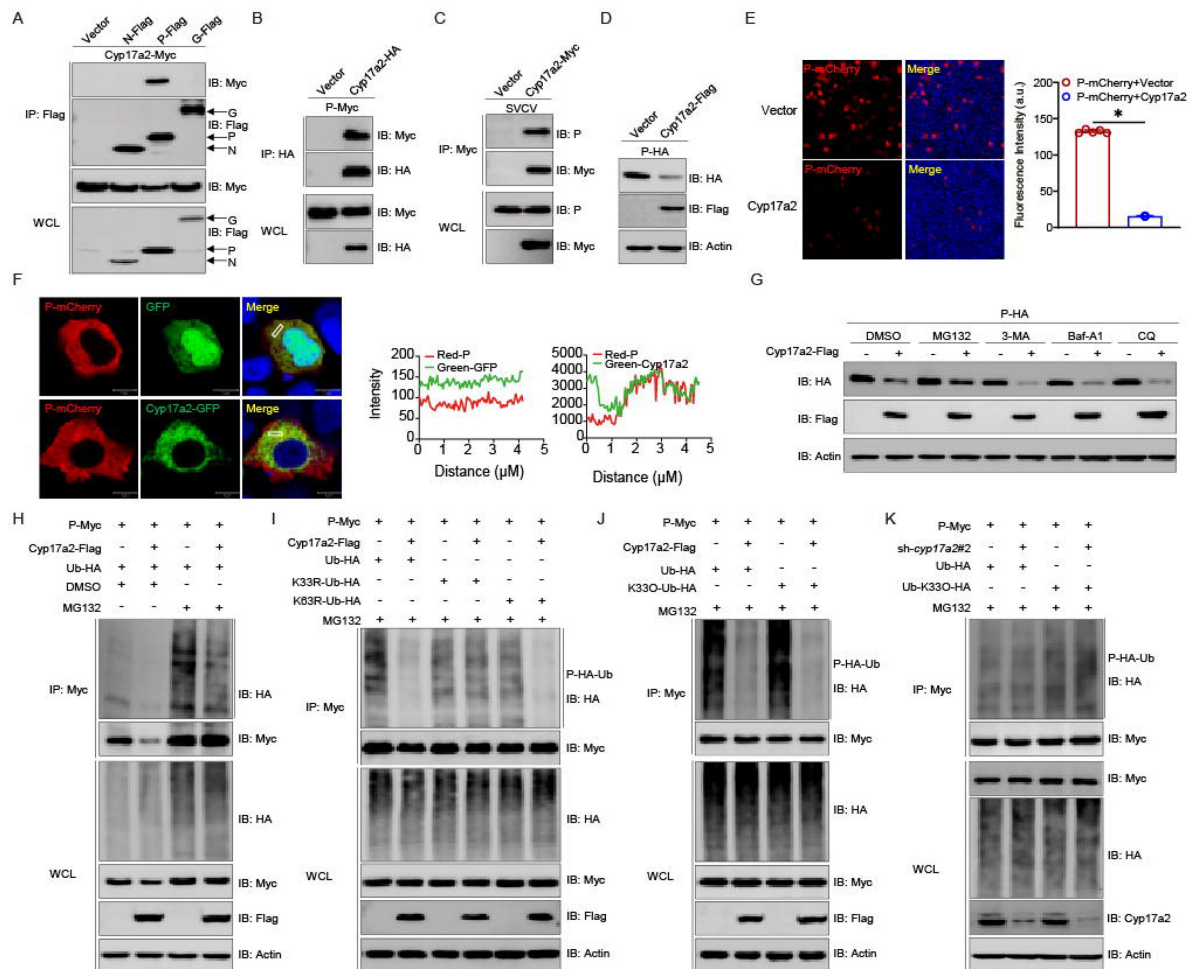


Figure 8

Cyp17a2 induces proteasomal degradation of the SVCV P protein by catalyzing K33-linked polyubiquitination.

(A-C) IB analysis of WCLs and proteins immunoprecipitated with anti-Flag, anti-HA, or anti-Myc Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (D) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (E) Fluorescent analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (F) Confocal microscopy of P and Cyp17a2 in EPC cells transfected with indicated plasmids for 24 h. The coefficient of colocalization was determined by qualitative analysis of the fluorescence intensity of the selected area in Merge. (G) IB analysis of proteins in EPC cells transfected with indicated plasmids for 18 h, followed by treatments of MG132 (10 μ M), 3-MA (2 mM), Baf-A1 (100 nM), and CQ (50 μ M) for 6 h, respectively. (H-K) P protein ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h.

9. The K12 site of P protein is essential for Cyp17a2-mediated degradation

To investigate the specific molecular mechanism underlying ubiquitination-mediated degradation of the P protein, mass spectrometry analysis revealed ubiquitin-specific protease 7 (USP7) and USP8 as candidate deubiquitinating enzymes (**Fig. 9A**). Both USP7 and USP8 were found to associate with the P protein, however, only USP8 interacted with Cyp17a2. Therefore, USP8 became the focus of subsequent study (**Fig. 9B** and 9C). The validity of these interactions was further confirmed through reciprocal experiments (**Fig. S8A**). Confocal microscopy analysis of USP8 subcellular localization demonstrated overlapping cytoplasmic co-localization between USP8-GFP (green fluorescence) and mCherry-tagged Cyp17a2 or the P protein (red fluorescence) (**Fig. 9D** and S8B). Cyp17a2 overexpression enhanced USP8 and P protein binding, whereas *cyp17a2* knockdown disrupted this interaction (**Fig. 9E** and S8C). USP8 overexpression amplified Cyp17a2-mediated P protein reduction (**Fig. 9F**). Knockdown of USP8 abolished Cyp17a2-dependent P protein degradation (**Fig. 9G** and S8D). USP8 facilitated Cyp17a2-induced deubiquitination of P protein, with knockdown reversing this effect (**Fig. 9H** and S8E). This result was also observed in the K33-linked ubiquitination assay (**Fig. 9I** and S8F). Mass spectrometry identified K12 as a critical lysine modification site (**Fig. 9J**). The K12R mutant resisted Cyp17a2-mediated degradation and showed impaired deubiquitination in WT ubiquitin and K33-linked assays (**Fig. 9K-9L** and S8G). USP8 failed to enhance Cyp17a2-mediated deubiquitination in K12R mutants (**Fig. 9M** and S8H). Truncation analysis demonstrated that USP8 residues 1-301 aa or 734-1067 aa were essential for interactions with P protein and Cyp17a2 (**Fig. 9N** and 9O). USP8 mutants only residues 1-301 aa or 302-733 aa lost the ability to support Cyp17a2-mediated P protein deubiquitination (**Fig. 9P** and S8I). These data establish that Cyp17a2 requires USP8 and the P protein K12 site to mediate deubiquitination-dependent degradation, with USP8 residues 734-1067 aa being indispensable for this process.

Discussion

The influence of biological sex on antiviral immunity shows both evolutionary parallels and divergences across species. While mammals typically exhibit stronger female immunity through sex chromosome and hormonal mechanisms, our study reveals an intriguing reversal in zebrafish, where males show enhanced antiviral responses via an autosomal gene-mediated pathway. This supplements conventional views and highlights the evolutionary plasticity of teleost immune systems.

Teleost display remarkable diversity in sex determination mechanisms, including XX/XY, ZZ/ZW, and environmental systems^{21, 22, 23}. This diversity provides a unique framework for studying sexual dimorphism independent of sex chromosomes. Using zebrafish, which lack conserved sex chromosomes, allows us to exclude the confounding effects of divergent sex-chromosome gene dosage and focus on the role of autosomal genes and their differential regulation in shaping sex-biased immunity. Our study leverages this unique context to demonstrate that enhanced antiviral immunity in males is driven not by canonical sex chromosomes or hormones, but by the male-biased expression of the autosomal gene *cyp17a2*.

Mechanistically, Cyp17a2 operates through dual antiviral pathways, it stabilizes STING via btr32-mediated K33-linked ubiquitination, enhancing IFN production, and recruits USP8 to remove K33-linked chains from viral P protein, promoting its degradation (**Fig. 10**). This establishes Cyp17a2 as a central regulator of sex-specific antiviral immunity independent of sex chromosome pathways. Particularly noteworthy is the demonstration that sexual dimorphism in immunity can

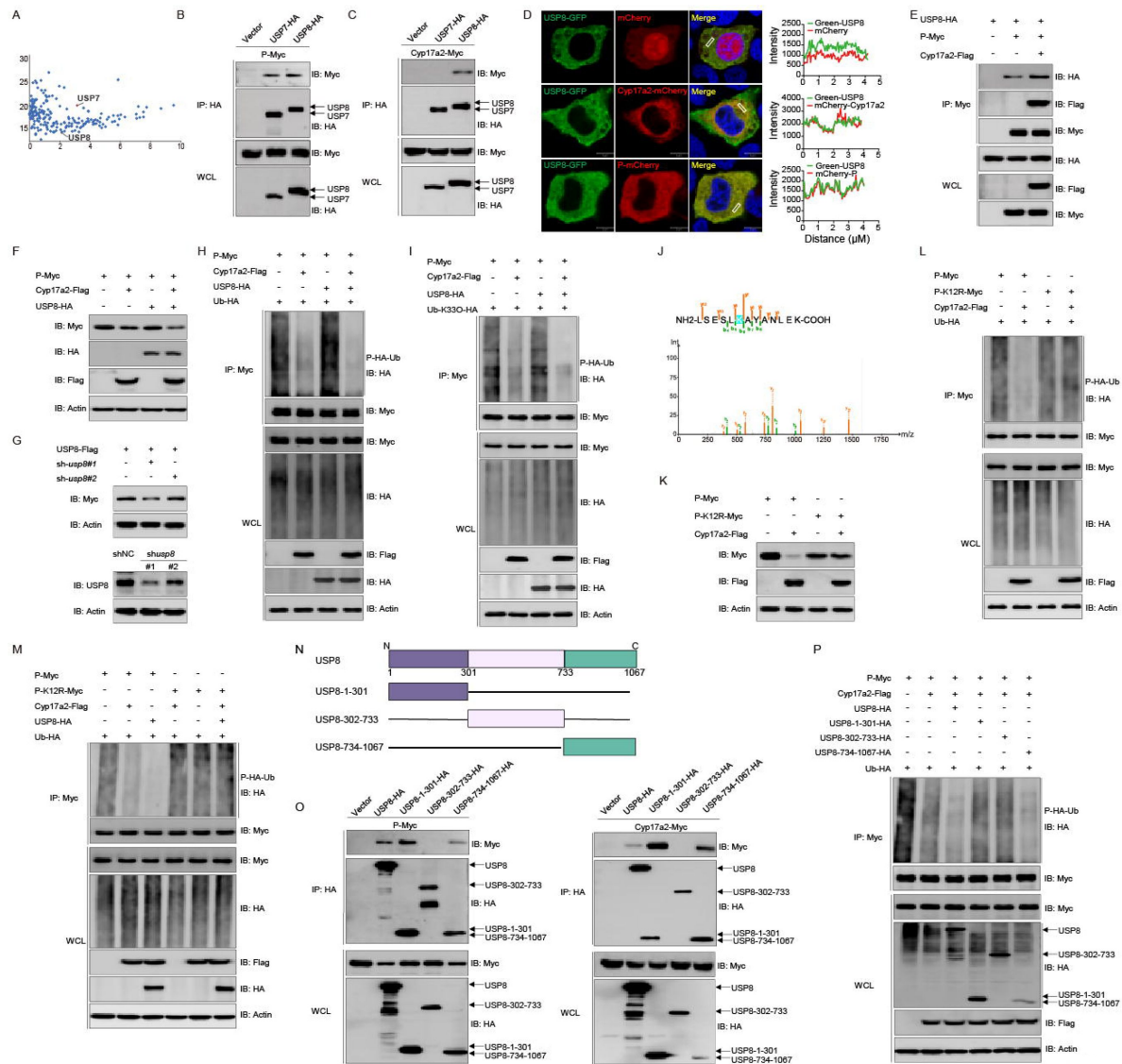


Figure 9

Cyp17a2 targets the K12 site of the P protein for K33-linked polyubiquitination.

(A) List of proteins interacting with P protein detected by mass spectrometry. (B and C, E, O) IB analysis of WCLs and proteins immunoprecipitated with anti-HA/MyC Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (D) Confocal microscopy of USP8 and Cyp17a2 or P protein in EPC cells transfected with indicated plasmids for 24 h. The coefficient of colocalization was determined by qualitative analysis of the fluorescence intensity of the selected area in Merge. (F and G, K) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (H and I, L and M, P) P protein ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h. (J) Mass spectrometry analysis of a peptide derived from ubiquitinated P-MyC. (N) Schematic representation of full-length USP8 and its mutants.

arise through differential regulation of autosomal genes rather than direct sex chromosome effects - a mechanism distinct from mammalian X-linked immune gene dosage effects or estrogen-mediated immunomodulation.

A key question emerging from our findings is whether the Cyp17a2-mediated antiviral mechanism is conserved across the teleost lineage or represents an adaptation specific to cyprinids such as zebrafish. The teleost-specific nature of *cyp17a2* suggests functions shaped by aquatic environmental challenges. While further studies in diverse teleost families are needed, the presence of *cyp17a2* orthologs across fish genomes indicates potential conservation. It is proposed that the recruitment of a core component of the steroidogenic enzyme machinery into the innate immune network may represent a significant evolutionary adaptation in teleost. Aquatic environments have been shown to have a persistently high pathogen load. The co-option of a gene such as *cyp17a2*, which is already subject to sex-biased regulatory control, offers an efficient and rapidly deployable strategy for establishing sexually dimorphic immunity, obviating the necessity for de novo evolution of entirely new pathways.

Sexual dimorphism in viral susceptibility represents a conserved yet mechanistically complex phenomenon across vertebrates²⁴. While males generally exhibit increased vulnerability to numerous viral pathogens such as dengue virus and influenza A virus, notable exceptions exist where females demonstrate higher susceptibility, including infections by herpes simplex virus and measles virus². These divergent patterns underscore a dual dependency of sex-biased infection outcomes on both host immunological determinants and viral evolutionary strategies¹. For instance, the male predominance observed in severe dengue virus cases may be explained by both gender-specific occupational exposure risks and sexual dimorphism in Ab- dependent enhancement mechanisms, potentially linked to androgen-mediated modulation of antiviral immune responses²⁵. Whereas female susceptibility to herpes simplex virus could be related to ovarian sex hormones, including estradiol and progesterone²⁶. Critically, sex differences in antiviral immunity are not static but evolve across the lifespan and are tightly linked to age and reproductive status²⁷. For example, sex disparities in hantavirus infection outcomes emerge post-puberty, coinciding with hormonal maturation²⁸. Similarly, during the 2009 H1N1 pandemic in Japan, morbidity rates exhibited a triphasic sex bias: males <20 and >80 years old showed higher susceptibility than females, whereas females of reproductive age (20-49 years) faced elevated risks²⁹. This pattern aligns with the hypothesis that estrogen peaks during reproductive years enhance antiviral defenses in females, while age-related declines in gonadal hormones may diminish this protective effect in postmenopausal populations³⁰. Our study extends these principles to teleost species, revealing novel sex-specific resistance patterns in fish-virus interactions. Male zebrafish demonstrated heightened resistance to SVCV, while male crucian carp exhibited reduced susceptibility to CyHV-2₃₁. These findings suggest that male-biased resistance may represent an evolutionarily conserved strategy in certain fish-virus system. This may reflect evolutionary adaptations to unique aquatic pressures. We propose several non-mutually exclusive hypotheses, which are rooted in the unique life histories and ecological pressures faced by aquatic vertebrates³². Firstly, the aquatic environment acts as a potent reservoir for pathogens, constantly exposing fish to diverse microbial challenges. This high pathogen pressure may select for specialized and potent immune strategies. If males and females encounter different levels or types of parasites or infections due to their behavior or morphology, this could lead to the evolution of sex-specific immune adaptations. Secondly, this phenomenon can be viewed through the lens of life-history trade-offs. Investment in immunity is costly in terms of energy and must be balanced against other demands, such as reproduction. In many teleost species, male reproductive success depends heavily on courtship displays, nest guarding and aggressive competition with other males, all of which are high-energy activities that can compromise immune function. Finally, the prevalence of external fertilization in teleost is a crucial factor. Unlike mammals, where the embryo develops internally and is protected by the maternal immune system, fish embryos are exposed to the pathogen-rich aquatic environment from the moment of fertilization. Therefore, a male's contribution to offspring survival includes

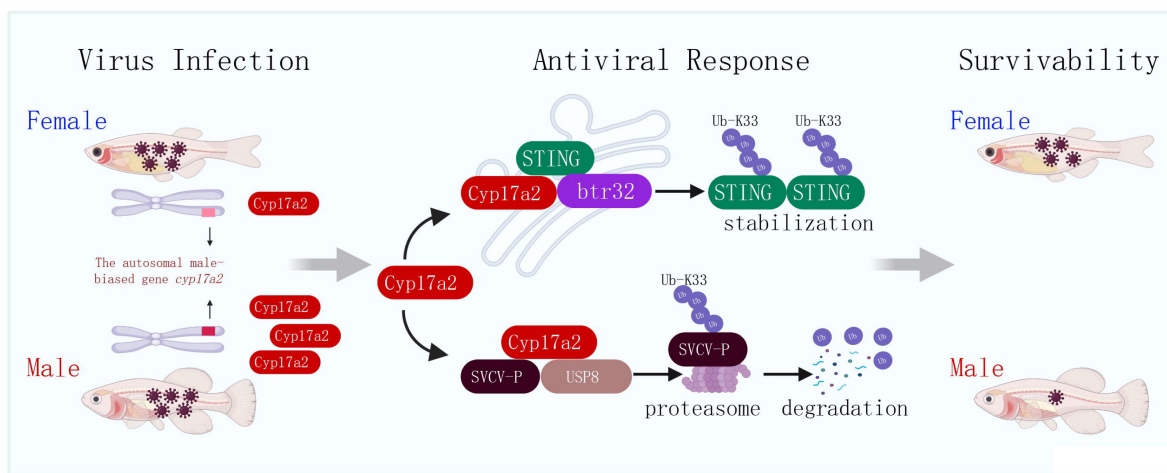


Figure 10

A mechanistic model illustrating the dual regulatory roles of Cyp17a2 in upregulating STING expression and degrading SVCV P protein.

Upon virus infection, male-biased autosomal gene *cyp17a2* enhances antiviral responses in males. Mechanistically, Cyp17a2, an ER-localized protein, stabilizes STING expression through recruitment of the E3 ubiquitin ligase btr32, promoting K33-linked polyubiquitination. Furthermore, Cyp17a2 facilitates proteasomal degradation of the SVCV P protein by engaging USP8 to reduce its K33-linked polyubiquitination, thereby amplifying antiviral IFN responses.

not only his genetic material, but also the intrinsic pathogen resistance of his sperm and the antimicrobial properties of the seminal fluid or nest site. STING was originally identified as a critical adaptor molecule mediating antiviral signaling cascades^{33, 34}. Subsequent studies have established its indispensable role in orchestrating innate immune responses against both DNA and RNA viral pathogens. Current mechanistic understanding reveals that upon detection of cytoplasmic pathogen-derived DNA, STING operates through two distinct molecular pathways: serving as a downstream effector of essential single-stranded/double-stranded DNA sensors or functioning as a direct receptor for microbial cyclic dinucleotides (CDNs). Following activation, this TM protein recruits and activates TBK1, which subsequently phosphorylates IRF3 to facilitate its nuclear translocation and initiate IFN production. Notably, the protein demonstrates functional pleiotropy in RNA viral infections through selective interaction with RIG-I rather than melanoma differentiation-associated protein 5 (MDA5), thereby activating parallel IRF3-dependent antiviral signaling pathways. Numerous studies indicate remarkable conservation of STING-mediated IFN induction mechanisms across teleost species^{35, 36, 37}. Mechanism studies further reveal that fish STING maintains dual antiviral functionality against both nucleic acid virus types, while simultaneously being subjected to targeted manipulation by various viral pathogens as an immune evasion strategy.

Beyond its steroidogenic functions, Cyp17a2 represents a novel scaffold protein in ubiquitination pathways. Despite the absence of clearly identifiable domains that are directly implicated in ubiquitination (e.g., RING, HECT, or U-box domains), it is proposed that the subcellular localization and protein-interaction capability of Cyp17a2 are pivotal to its novel function. As an ER-resident protein, Cyp17a2 is strategically co-localized with central innate immune sensors such as STING. This shared localization facilitates specific protein-protein interactions, enabling Cyp17a2 to serve as an essential platform that recruits the E3 ubiquitin ligase btr32 to stabilize STING via K33-linked ubiquitination. In contrast, Cyp17a2 engages the deubiquitinate USP8 to remove K33-linked chains from the viral P protein, thereby targeting it for degradation. This bifunctional and scaffolding role, which orchestrates both the addition and removal of specific ubiquitin chains to fine-tune antiviral responses, represents a fascinating evolutionary adaptation in teleost.

The CYP450 enzyme superfamily exhibits multifaceted roles across species, encompassing physiological processes and immune regulation. In human immune cells, pharmacological inhibition of CYP1A has been demonstrated to upregulate the expression of stem cell factor receptor and interleukin 22, directly linking CYP1-dependent metabolism of environmental small molecules to immune modulation³⁸. Evolutionary conservation of this regulatory mechanism is evident in teleost, where grass carp (*Ctenopharyngodon idella*) CYP1A demonstrates dynamic expression during grass carp reovirus (GCRV) infection, implicating its role in antiviral immunity³⁹. Our study further identifies Cyp17a2 as a sexually dimorphic regulator of IFN signaling in fish. This finding expands the recognized roles of the CYP450 superfamily in immune regulation, revealing a novel mechanism underlying sex-specific immunity in lower vertebrates.

In conclusion, zebrafish model with its lack of conserved sex chromosomes and environmental sex determination plasticity, provides unique insights into the evolutionary continuum of sexual dimorphism. Our results suggest that the relationship between sexual differentiation and immune competence may be more fluid in teleost than in mammals, potentially reflecting adaptive responses to aquatic pathogens and variable mating systems. Future investigations should explore whether this autosomal-centered mechanism represents a teleost specific adaptation or an ancestral feature subsequently modified in terrestrial vertebrates, while also examining potential interactions between environmental sex determinants and immune gene regulation.

STAR Methods

Ethics statement

The experiments involved in this study were conducted in compliance with ethical regulations. The fish experiments were carried out under the guidance of the European Union Guidelines for the Handling of Laboratory Animals (2010/63/EU) and approved by the Ethics Committee for Animal Experiments of the Institute of Hydrobiology, Chinese Academy of Sciences (CAS) (No. 2024-069).

Fish, cells, and viruses

Mature zebrafish individuals 2.5 months after hatching (0.4 ± 0.1 g) were selected in this study. Wild-type (WT) zebrafish (AB strain; *Danio rerio*) were procured from the China Zebrafish Resource Center (CZRC), while *cyp17a2* mutant lines were generously provided by Dr. Zhan Yin's laboratory at the Institute of Hydrobiology, Chinese Academy of Sciences. All specimens were bred using standardized procedures. In accordance with the ethical requirements and national animal welfare guidelines, all experimental fish were required to undergo a two-week acclimatization period in the laboratory and have their health assessed prior to the commencement of the study. Only fish that appeared healthy and were mobile were selected for scientific investigation. ZF4 cells (American Type Culture Collection, ATCC) were cultured in Ham's F-12 medium (Invitrogen) supplemented with 10% fetal bovine serum (FBS) at 28°C and 5% CO₂. EPC cells were obtained from the Chinese Culture Collection Centre for Type Cultures (CCTCC), Gibel carp brain (GiCB) cells were provided by Ling-Bing Zeng (Yangtze River Fisheries Research Institute, Chinese Academy of Fishery Sciences), these cells were maintained at 28°C in 5% CO₂ in medium 199 (Invitrogen) supplemented with 10% FBS. Spring viremia of carp virus (SVCV) was propagated in EPC cells until a CPE was observed, and then cell culture fluid containing SVCV was harvested and centrifuged at 4×10^3 g for 20 min to remove the cell debris, and the supernatant was stored at -80°C until used. Cyprinid herpesvirus 2 (CyHV-2, obtained from Yancheng city, Jiangsu province, China) was provided by Prof. Liqun Lu (Shanghai Ocean University). CyHV-2 was propagated in GiCB cells and harvested in a similar way to SVCV.

Plasmid construction and reagents

The sequences of zebrafish and gibel carp (*Carassius gibelio*) *Cyp17a2* (GenBank accession number: NM_001105670.1 and XM_026199804.1) were obtained from the National Centre for Biotechnology Information (NCBI) website. *Cyp17a2* was amplified by polymerase chain reaction (PCR) using cDNA from adult zebrafish or gibel carp tissues as a template and cloned into the expression vector pCMV-HA or pCMV-Myc (Clontech) vectors. Zebrafish MAVS (NM_001080584.2), TBK1 (NM_001044748.2), STING (NM_001278837.1) and the truncated mutants of STING, IRF3 (NM_001143904.1), IRF7 (BC058298.1), btr32 (XM_021470074.1) and the truncated mutants of btr32, TRAP1 (NM_205607.1), SOCS3a (NM_199950.1), USP7 (XM_021473871.1), USP8 (XM_009293353.4), and N, P, and G protein (DQ097384.2) of SVCV were cloned into pCMV-Myc and pCMV-Tag2C vectors. The shRNA of *Pimephales promelas* *Cyp17a2* (XM_039679078.1), STING (HE856620.1), and USP8 (XM_039680290.1) were designed by BLOCK-iT RNAi Designer and cloned into the pLKO.1-TRC Cloning vector. For subcellular localization experiments, *Cyp17a2* was constructed onto pEGFP-N3 (Clontech), while STING and P protein were constructed onto pCS2-mCherry (Clontech). The plasmids containing zebrafish IFN β 1pro-Luc and ISRE-Luc in the pGL3-Basic luciferase reporter vector (Promega) were constructed as described previously. The *Renilla* luciferase internal control vector (pRL-TK) was purchased from Promega. The ubiquitin mutant expression plasmids Lys-33/63 (all lysine residues were mutated except Lys-33 or Lys-63) and Lys-33R/63R (only Lys-33 or Lys-63 mutated) were ligated into the pCMV-HA vectors named Ub-K33O/K63O-HA and Ub-K33R/K63R. All constructs were confirmed by DNA sequencing. Polyinosinic-polycytidylic

acid (poly I:C) was purchased from Sigma-Aldrich used at a final concentration of 1 µg/µl. MG132 (Cat. No. M7449), 3-Methyladenine (3-MA) (Cat. No. M9281), CQ (Cat. No. C6628) were obtained from Sigma-Aldrich. Bafilomycin A1 (Baf-A1) (Cat. No. S1413) and CHX (NSC-185) were obtained from Selleck.

Transcriptomic analysis

Total RNA was extracted using the TRIzol method and evaluated for RNA purity and quantification using a NanoDrop 2000 spectrophotometer (Thermo Scientific, Waltham, U.S.A.) and RNA integrity was assessed using an Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, U.S.A.). Transcriptome sequencing and data analysis were conducted by OE Biotech (Shanghai, China). The raw sequencing data was submitted to the NCBI (GEO accession number: GSE286486).

Transient transfection and virus infection

EPC cells were transfected in 6-well and 24-well plates using transfection reagents from FishTrans (MeiSenTe Biotechnology) according to the manufacturer's protocol. Antiviral assays were conducted in 24-well plates by transfecting EPC cells with the plasmids shown in the figure. At 24 h post-transfection, cells were infected with SVCV at a multiplicity of infection (MOI = 0.001). After 24 h, supernatant aliquots were harvested for detection of virus titers, the cell monolayers were fixed by 4% paraformaldehyde (PFA) and stained with 1% crystal violet for visualizing CPE. For virus titration, 200 µl of culture medium were collected at 48 h post-infection and used for detection of virus titers according to the method of Reed and Muench. The supernatants were subjected to 3-fold or 10-fold serial dilutions and then added (100 µl) onto a monolayer of EPC cells cultured in a 96-well plate. After 48 or 72 h, the medium was removed and the cells were washed with phosphate-buffered saline (PBS), fixed by 4% PFA, and stained with 1% crystal violet. The virus titer was expressed as 50% tissue culture infective dose (TCID₅₀/ml). For viral infection, fish were anesthetized with methanesulfonate (MS-222) and intraperitoneally (i.p.) injected with 5 µl of M199 containing SVCV (5×10^8 TCID₅₀/ml). The i.p. injection of PBS was used as mock infection. Then the fish were migrated into the aquarium containing new aquatic water.

Luciferase activity assay

EPC cells were cultured overnight in 24-well plates and subsequently co-transfected with the expression plasmid and luciferase reporter plasmid. The cells were infected with SVCV or transfected with poly I:C for 24 h prior to harvest. At 24 h post-stimulation, cells were washed with PBS and lysed for measuring luciferase activity by the Dual-Luciferase Reporter Assay System (Promega) according to the manufacturer's instructions. Firefly luciferase activity was normalized based on the *Renilla* luciferase activity.

RNA extraction, reverse transcription, and quantitative PCR (qPCR)

The RNA was extracted using TRIzol reagent (Invitrogen), and first-strand cDNA was synthesized with a PrimeScript RT kit with gDNA Eraser (Takara). qPCR was performed on the CFX96 Real-Time System (Bio-Rad) using SYBR green PCR Master Mix (Yeast). The PCR conditions were as follows: 95°C for 5 min and then 40 cycles of 95°C for 20 s, 60°C for 20 s, and 72°C for 20 s. The primers utilized for the qPCRs are presented in Table S1, and the *β-actin* gene was utilized as the internal control. The relative fold changes were calculated by comparison to the corresponding controls using the $2^{-\Delta\Delta C_t}$ method (where C_t was the threshold cycle).

Co-immunoprecipitation (Co-IP) assay

EPC cells were cultured in 10-cm² dishes overnight and subsequently transfected with 10 µg plasmid as illustrated. At 24 h post-transfection, the medium was removed and cells were washed with PBS. Then the cells were lysed in 1 ml radioimmunoprecipitation (RIPA) lysis buffer [1% NP-40, 50 mM Tris-HCl, pH 7.5, 150 mM NaCl, 1 mM EDTA, 1 mM NaF, 1 mM sodium orthovanadate

(Na_3VO_4), 1 mM phenyl-methylsulfonyl fluoride (PMSF), 0.25% sodium deoxycholate] containing protease inhibitor cocktail (Sigma-Aldrich) at 4°C for 1 h on a rocker platform. The cellular debris was removed by centrifugation at $12,000 \times g$ for 15 min at 4°C. The supernatant was transferred to a fresh tube and incubated with 20 μL anti-Flag/HA/Myc affinity gel (Sigma-Aldrich) overnight at 4°C with constant rotating incubation. These samples were further analyzed by immunoblotting (IB). Immunoprecipitated proteins were collected by centrifugation at $5000 \times g$ for 1 min at 4°C, washed three times with lysis buffer and resuspended in 50 μL 2 $\times$ SDS sample buffer. The immunoprecipitates and whole cell lysates (WCLs) were analyzed by IB with the indicated antibodies (Abs). ***In vivo* ubiquitination assay**

The transfected EPC cells were washed twice with 10 mL ice-cold PBS and subsequently digested with 1 mL 0.25% trypsin-EDTA (1 $\times$) (Invitrogen) for 2-3 min until the cells were dislodged. 100 μL FBS was added to neutralize the trypsin and the cells were resuspended into 1.5 mL centrifuge tube, centrifuged at $2000 \times g$ for 5 min. The supernatant was discarded and the cell precipitations were resuspended using 1 mL PBS and centrifuged at $2000 \times g$ for 5 min. The collected cell precipitations were lysed using 100 μL PBS containing 1% SDS and denatured by heating for 10 min. The supernatants were diluted with lysis buffer until the concentration of SDS was reduced to 0.1%. The diluted supernatants were incubated with 20 μL anti-Myc affinity gel overnight at 4°C with constant agitation. Subsequently, these samples were subjected to further analysis by IB. The immunoprecipitated proteins were collected by centrifugation at $5000 \times g$ for 1 min at 4°C, washed for three times with lysis buffer and resuspended in 100 μL 1 $\times$ SDS sample buffer.

Immunoblot analysis

Immunoprecipitates or WCLs were analyzed as described previously. Abs were diluted as follows: anti- β -actin (ABclonal, AC026) at 1:10000, anti-Flag (Sigma-Aldrich, F1804) at 1:3000, anti-HA (Covance, MMS-101R) at 1:3000, anti-Myc (Santa Cruz Biotechnology, sc-40) at 1:3000, anti-btr32 (ABclonal, A13887) at 1:1000, anti-USP8 (proteintech, 27791-1-AP) at 1:1000, and HRP-conjugated anti-mouse/rabbit IgG (Thermo Scientific, 31430/31460) at 1:5000, anti-N/P/G/STING/ (prepared and purified in our lab) at 1:2000.

Immunofluorescence (IF)

EPC cells were plated onto glass coverslips in 6-well plates and infected with SVCV (MOI = 1) 24 h. The cells were then washed with PBS and fixed in 4% PFA at room temperature for 1 h and permeabilized with 0.2% Triton X-100 in ice-cold PBS for 15 min. The samples were incubated for 1 h at room temperature in PBS containing 2% bovine serum albumin (BSA, Sigma-Aldrich). After additional PBS washing, the samples were incubated with anti-N Ab in PBS containing 2% BSA for 2-4 h at room temperature. After three times washed by PBS, the samples were incubated with secondary Ab (Goat anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 488) (Thermo Scientific, A32731, 1:5000) in PBS containing 2% BSA for 1 h at room temperature. After additional PBS washing, the cells were finally stained with 1 $\mu\text{g}/\text{mL}$ 4', 6-diamidino-2-phenylindole (DAPI, Beyotime Institute of Biotechnology, Shanghai, China) for 10 min in the dark at room temperature. Finally, the coverslips were washed and observed with a confocal microscope under a 10 $\times$ immersion objective (SP8; Leica).

Fluorescent microscopy

EPC cells were plated onto coverslips in 6-well plates and transfected with the plasmids indicated for 24 h. Thereafter, the cells were washed twice with PBS and fixed with 4% PFA for 1 h. After being washed three times with PBS, the cells were stained with 1 $\mu\text{g}/\text{mL}$ DAPI for 15 min in the dark at room temperature. Subsequently, the coverslips were washed and observed with a confocal microscope under a 63 $\times$ oil immersion objective (SP8; Leica).

Histopathology

Liver and spleen tissues from three individuals of control or virus-infected fish at 2 days post infection (dpi) were dissected, and fixed in Bouin's Fixative overnight. Subsequently, the samples were dehydrated in ascending grades of alcohol and embedded into paraffin. Sections at 5 μ m thickness were taken and stained with hematoxylin and eosin (H&E). Histological changes were examined by optical microscopy at $\times 40$ magnification and were analyzed by the Aperio ImageScope software.

Statistics analysis

For fish survival analysis, Kaplan-Meier survival curves were generated and subsequently subjected to a log-rank test. For the bar graph, one representative experiment of at least three independent experiments is shown, and each was done in triplicate. For the dot plot graph, each dot point represents one independent biological replicate. Unpaired Student's *t* test was used for statistical analysis. Data are presented as mean $\pm$ standard error of the mean (SEM). A *p* value < 0.05 was considered statistically significant.

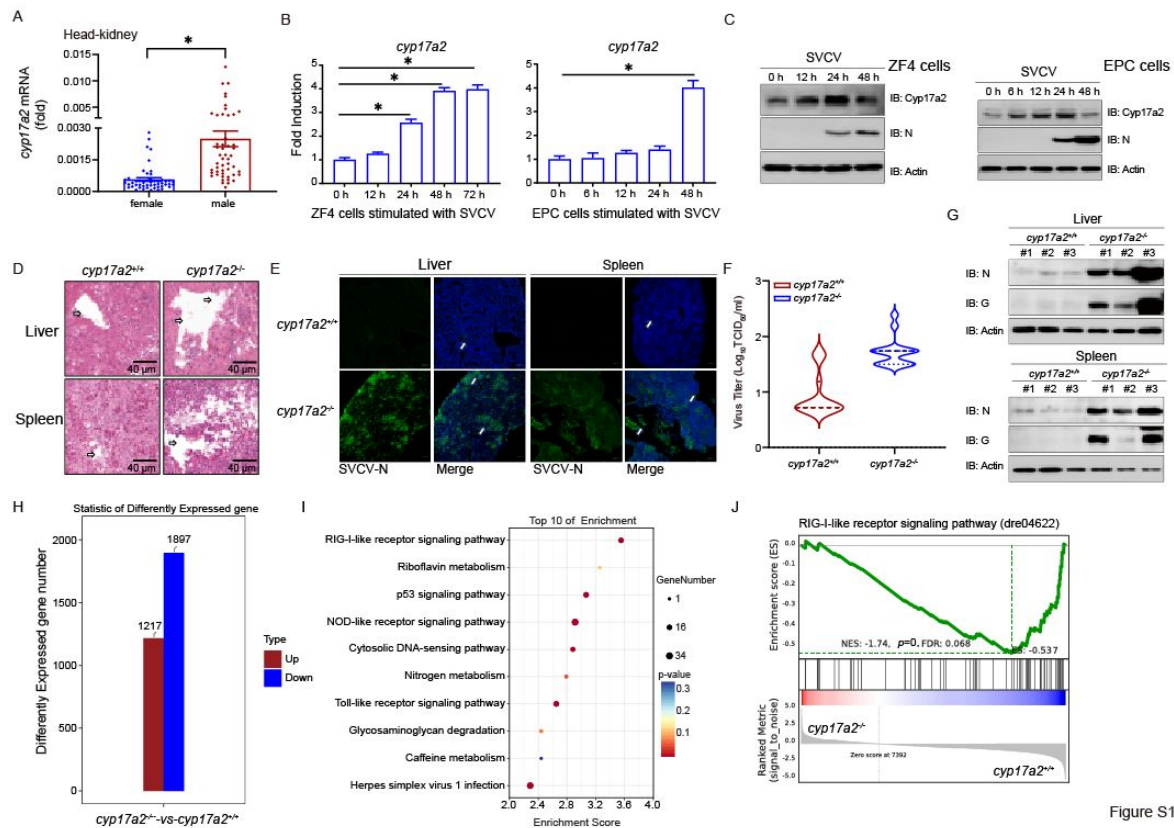


Figure S1

Figure S1

The male-biased expression of Cyp17a2 governs RLR antiviral signaling.

(A) qPCR analysis of *cyp17a2* mRNA in the head-kidney of male and female zebrafish (*n* = 50 per group). (B) qPCR analysis of *cyp17a2* mRNA in the ZF4 or EPC cells infected with SVCV for the indicated times. (C) IB analysis of Cyp17a2 in ZF4 and EPC cells infected with SVCV for the indicated times. (D) Microscopy of H&E-stained liver and spleen sections from *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish treated with SVCV for 48 h. (E) IF analysis of SVCV-N protein in the liver and spleen of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish treated with SVCV for 48 h. (F) The viral titer of heart in *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish (*n* = 15 per group) treated with SVCV for 48 h. (G) IB analysis of SVCV proteins in the liver and spleen sections of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish (*n* = 3 per group) treated with SVCV for 48 h. (H) The differently expressed gene number of mRNA variations in liver of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish infected with SVCV. (I) KEGG pathway enrichment map showing the top 10 signaling pathways of total. (J) GSEA of differentially expressed genes in the liver of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish infected with SVCV and enrichment of RLR signaling pathway. The *p*-values were corrected for the False Discovery Rate (FDR) method, and modules with *p*-values less than 0.05 were selected as significant.

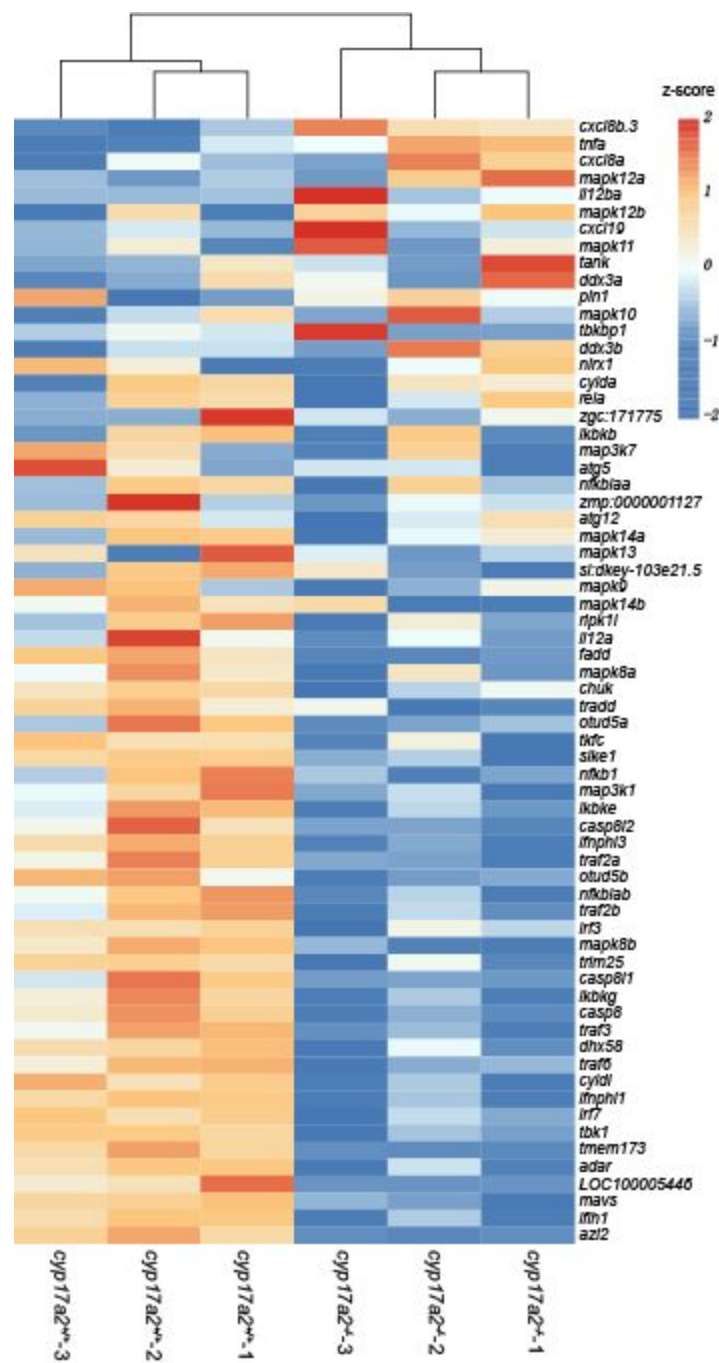


Figure S2

Figure S2

Heatmap view of mRNA variations of RLR-mediated ISGs sets in the liver of *cyp17a2*^{+/+} and *cyp17a2*^{-/-} male zebrafish infected with SVCV.

Expression values are $\log_2(\text{FPKM}+1)$ transformed and presented as row Z-scores.

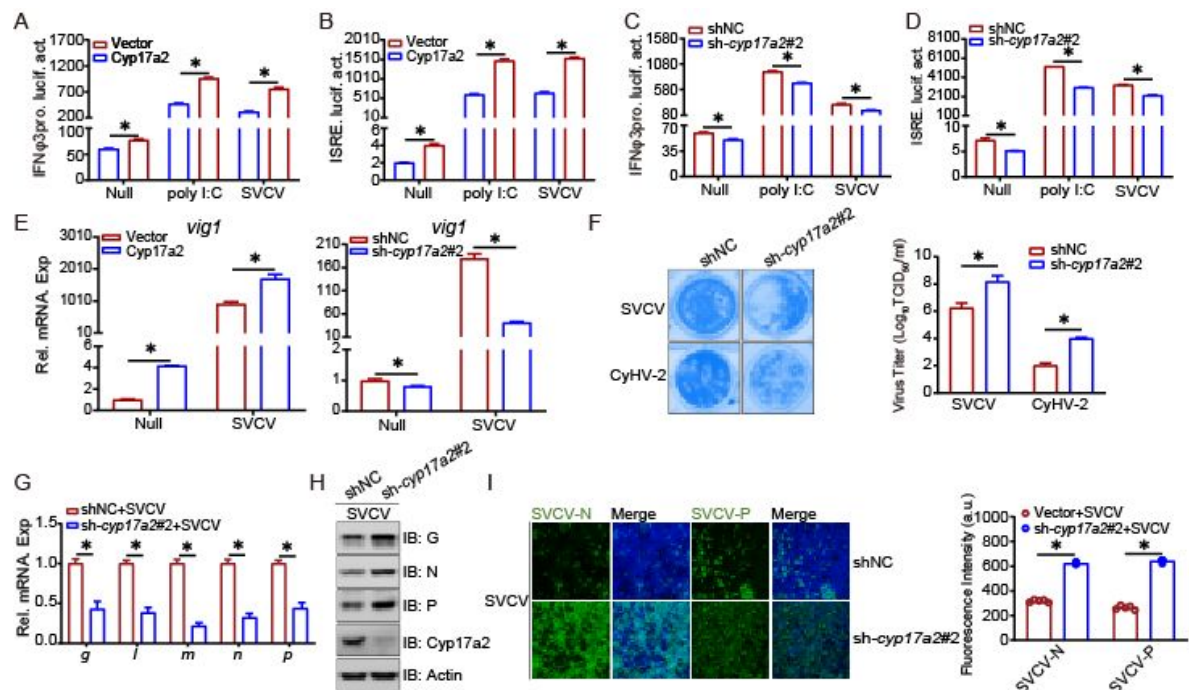


Figure S3

Figure S3

Cyp17a2 enhances IFN production to inhibit viral replication.

(A-D) Luciferase activity of IFN3pro and ISRE in EPC cells transfected with indicated plasmids for 24 h, and then untreated or transfected with poly I:C (0.5 µg) or infected with SVCV (MOI = 1) for 24 h before luciferase assays. (E) qPCR analysis of *vig1* in EPC cells transfected with indicated plasmids for 24 h, and then untreated or infected with SVCV (MOI = 1) for 24 h. (F) Plaque assay of virus titers in EPC cells or GICB cells transfected with indicated plasmids for 24 h, followed by SVCV or CyHV-2 challenge for 24 h or 48 h. (G and H) qPCR and IB analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. (I) IF analysis of SVCV proteins in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h.

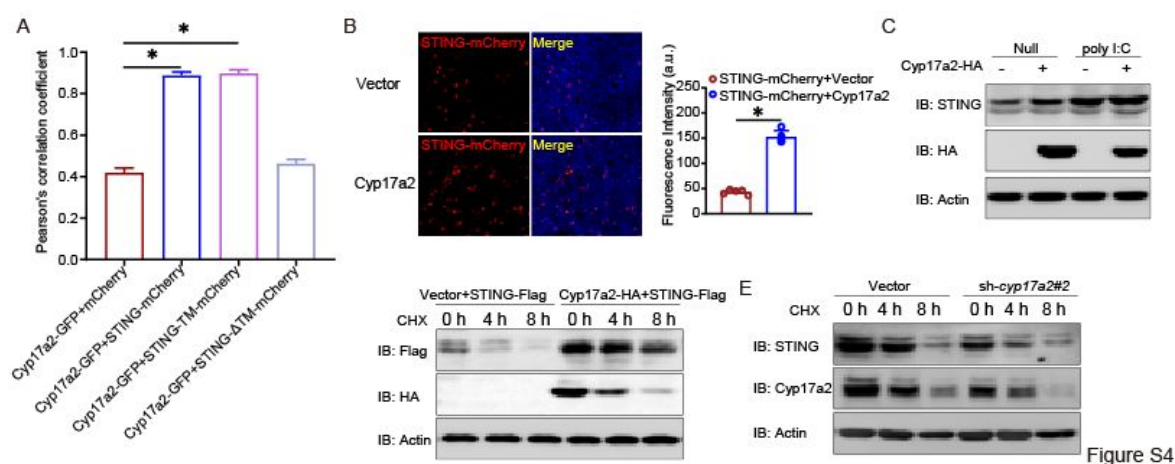


Figure S4

Figure S4

STING stability is enhanced by Cyp17a2.

(A) Colocalization analyses of **figure 4G** were performed by calculating Pearson correlation coefficient. (B) Fluorescent analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. The fluorescence intensity (arbitrary unit, a.u.) was recorded by the LAS X software, and the data were expressed as mean $\pm$ SD, $n = 5$. (C) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h, followed by untreated or transfected with poly I:C (2 μ g) for 24 h. (D and E) IB analysis of proteins in EPC cells transfected with indicated plasmids for 18 h, then treated with CHX for 4 h and 8 h.

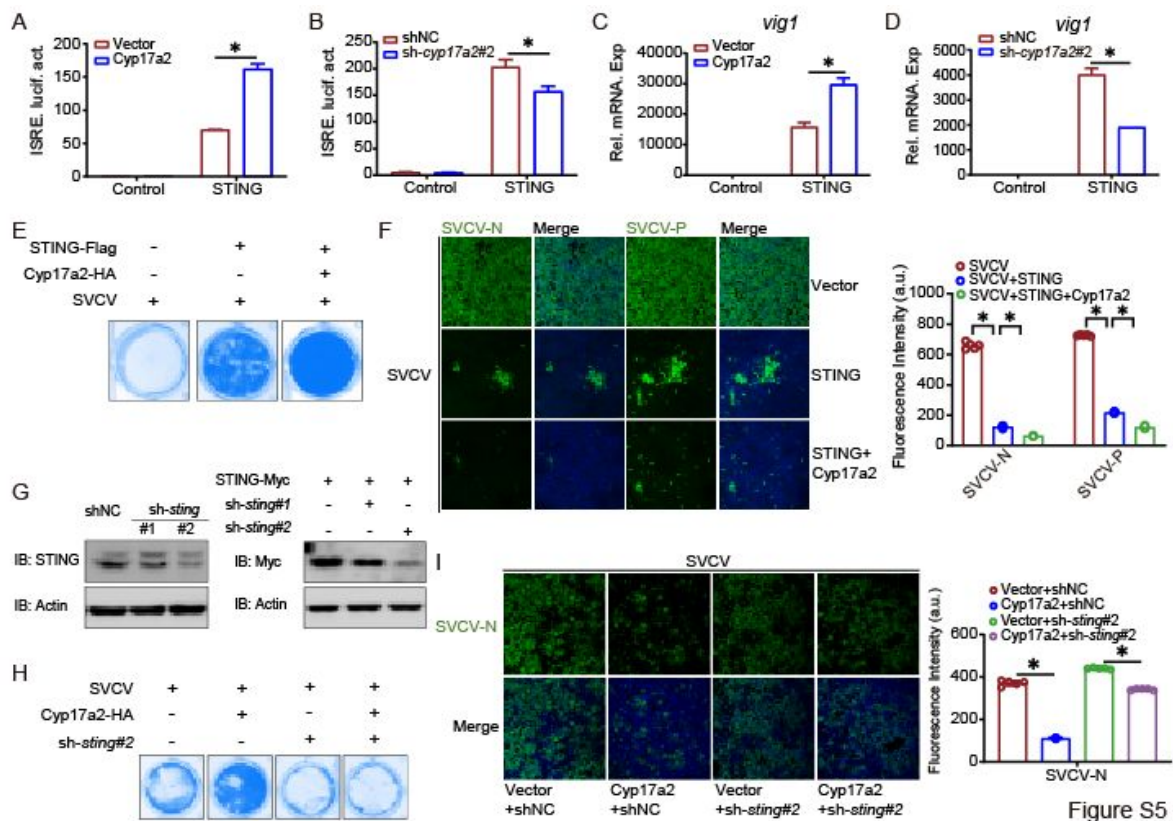


Figure S5

Cyp17a2 enhances STING-mediated IFN activation and antiviral defense.

(A and B) Luciferase activity of ISRE in EPC cells transfected with indicated plasmids for 24 h before luciferase assays. (C and D) qPCR analysis of *vig1* in EPC cells transfected with indicated plasmids for 24 h. (E and H) Plaque assay in EPC cells transfected with indicated plasmids for 24 h and then infected with SVCV (MOI = 1) for 24 h. (F and I) IF analysis of N protein and P protein in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. The fluorescence intensity (arbitrary unit, a.u.) was recorded by the LAS X software, and the data were expressed as mean $\pm$ SD, $n = 5$. (G) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h.

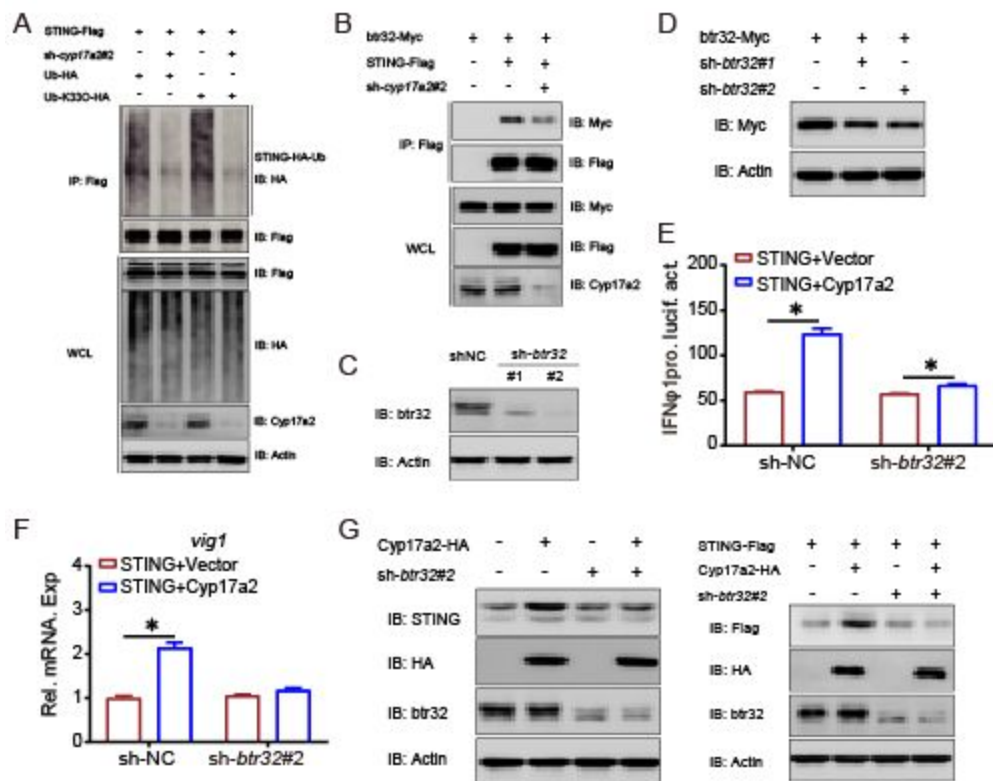


Figure S6

Figure S6

Cyp17a2 promotes STING stabilization by facilitating btr32-catalyzed K33-linked polyubiquitination.

(A) STING ubiquitination assays in EPC cells transfected with indicated plasmids for 24 h. (B) IB analysis of WCLs and proteins immunoprecipitated with anti-Flag Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (C and D, G) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (E and F) Luciferase activity of IFNβ1pro and qPCR analysis of *vig1* in EPC cells transfected with indicated plasmids for 24 h.

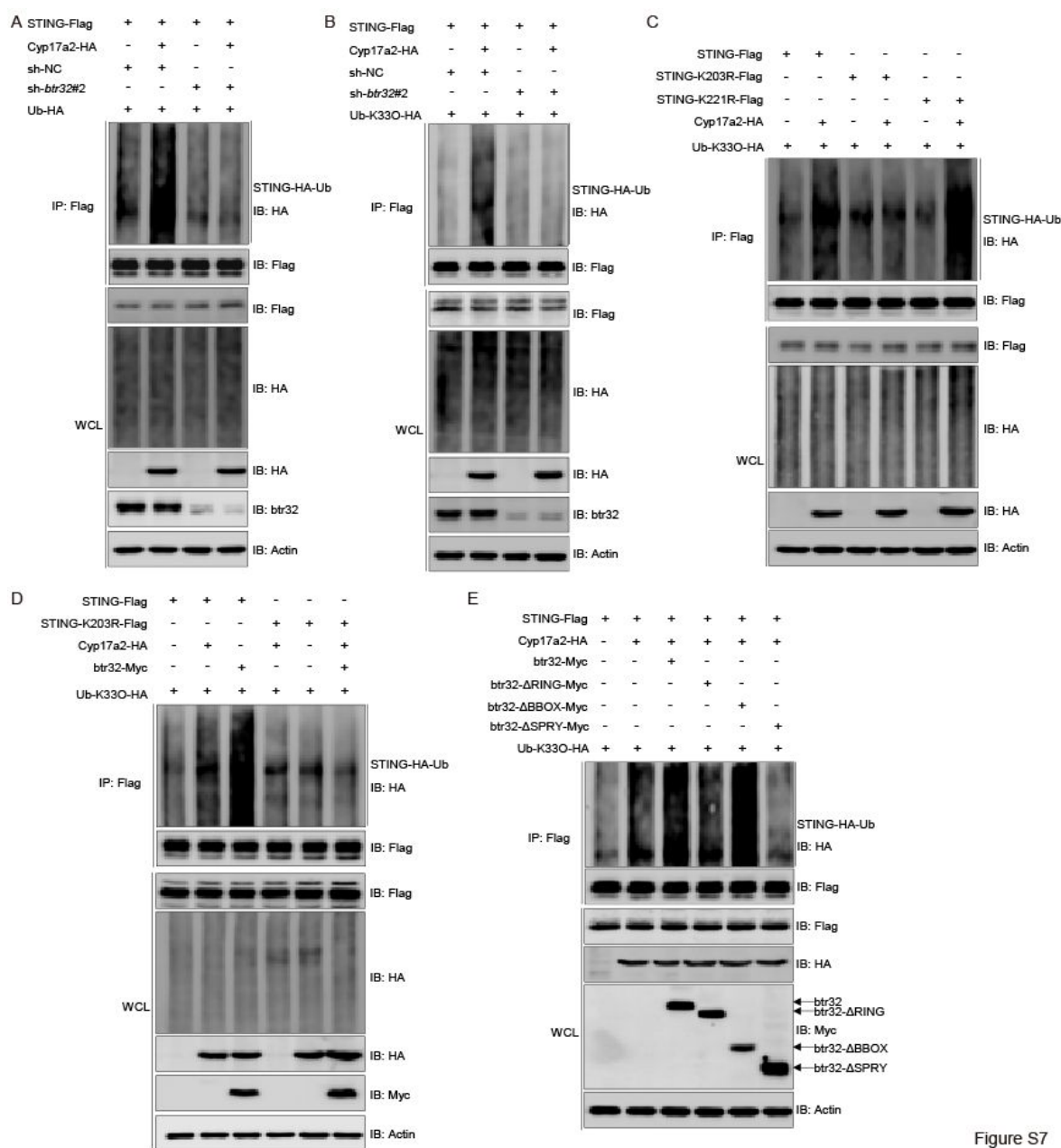


Figure S7

Figure S7

The K203 residue is essential for the ubiquitination of STING mediated by Cyp17a2.

(A-E) STING ubiquitination assays in EPC cells transfected with indicated plasmids for 24 h.

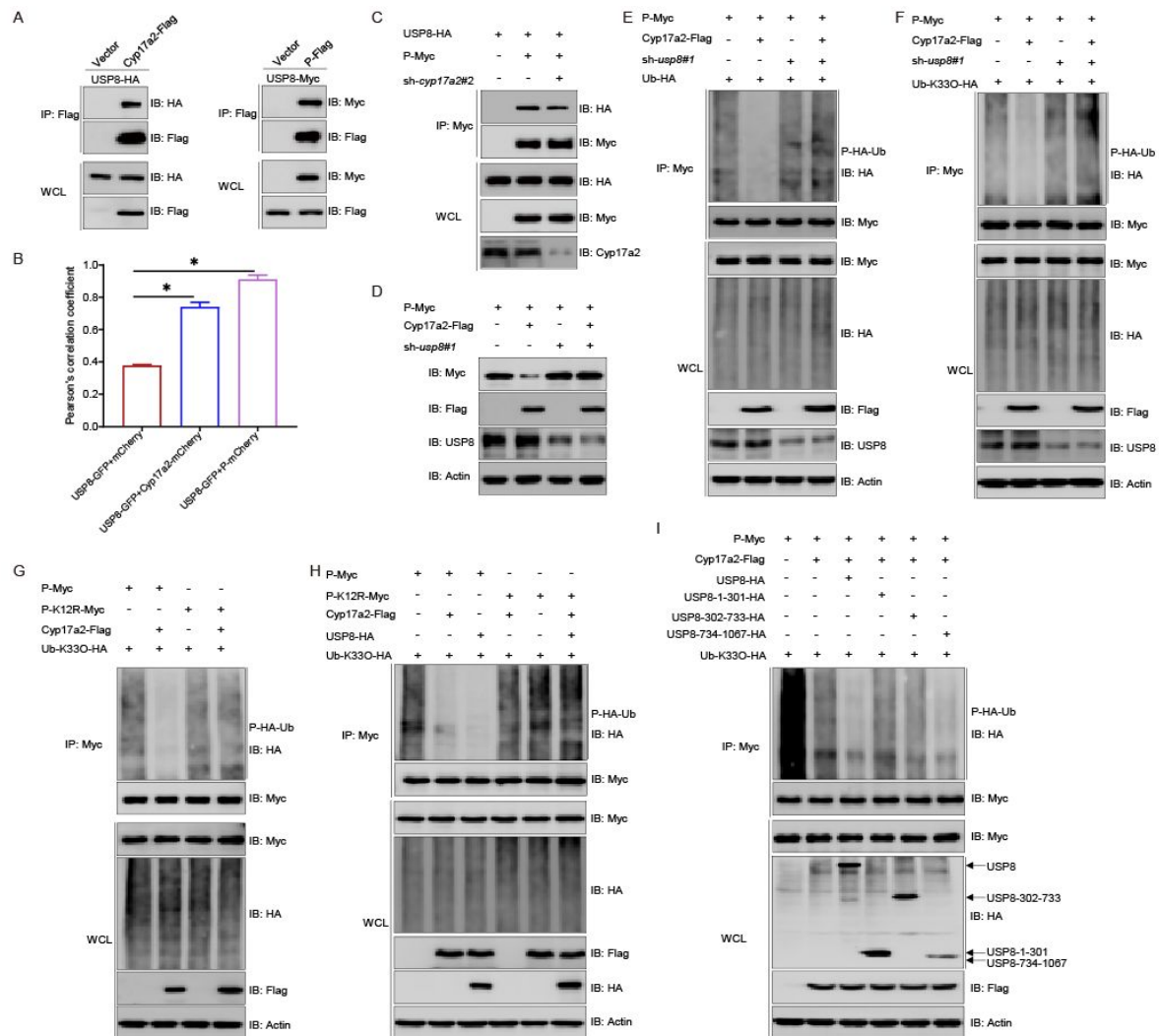


Figure S8

Figure S8

Cyp17a2 mediates K33-linked polyubiquitination of the P protein at K12.

(A and C) IB analysis of WCLs and proteins immunoprecipitated with anti-Flag/Myc Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (B) Colocalization analyses of [figure 9D](#) were performed by calculating Pearson correlation coefficient. (D) IB analysis of proteins in EPC cells transfected with indicated plasmids for 24 h. (E-I) P protein ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h.

Data availability

RNA sequence data were deposited in GEO (accession number: GSE286486). All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

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Additional files

Table S1. List of primer sequences used in this study. [↗](#)

Table S2. List of all identified genes in FHK-vs-MHK in transcriptomic analysis. [↗](#)

Table S3. Heatmap view of mRNA variations of SVCV-activated ISG sets in the *Cyp17a2*-overexpressing cells or *cyp17a2* knockdown cells. [↗](#)

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Reviewer #1 (Public review):

Summary:

In this manuscript Lu & Cui et al. observe that adult male zebrafish are more resistant to infection and disease following exposure to Spring Viremia of Carp Virus (SVCV) than female fish. The authors then attempt to identify some of the molecular underpinnings of this apparent sexual dimorphism and focus their investigations on a gene called cytochrome P450, family 17, subfamily A, polypeptide 2 (cyp17a2) because it was among genes that they found to be more highly expressed in kidney tissue from males than in females. Their investigations lead them to propose a direct connection between cyp17a2 and modulation of interferon signaling as the key underlying driver of difference between male and female susceptibility to SVCV.

Strengths:

Strengths of this study include the interesting observation of a substantial difference between adult male and female zebrafish in their susceptibility to SVCV, and also the breadth of experiments that were performed linking cyp17a2 to infection phenotypes and molecularly to the stability of host and virus proteins in cell lines. The authors place the infection phenotype in an interesting and complex context of many other sexual dimorphisms in infection phenotypes in vertebrates. This study succeeds in highlighting an unexpected factor involved in antiviral immunity that will be an important subject for future investigations of infection, metabolism, and other contexts.

Weaknesses:

Weaknesses of this study include a proposed mechanism underlying the sexual dimorphism phenotype based on experimentation in only males, and widespread reliance on over-expression when investigating protein-protein interaction and localization. Additionally, a minor weakness is that the text describing the identification of cyp17a2 as a candidate contains errors that are confusing. For example:

- Lines 139-140 describe the data for Figure 2 as deriving from "healthy hermaphroditic adult zebrafish". This appears to be a language error and should be corrected to something that specifies that the comparison made is between healthy adult male and female kidneys.
- In Figure 2A and associated text cyp17a2 is highlighted but the volcano plot does not indicate why this was an obvious choice. For example, many other genes are also highly induced in male vs female kidneys. Figure 2B and line 143 describe a subset of "eight sex-related genes" but it is not clear how these relate to Figure 2A. The narrative could be improved to clarify how cyp17a2 was selected from Figure 2A and it seems that the authors made an attempt to do this with Figure 2B but it is not clear how these are related. This is important because the available data do not rule out the possibility that other factors also mediate the sexual dimorphism they observed either in combination, in a redundant fashion, or in a more complex genetic fashion. The narrative of the text and title suggests that they consider this to be a monogenic trait but more evidence is needed.

<https://doi.org/10.7554/eLife.108048.2.sa2>

Reviewer #2 (Public review):

This study conducted by Lu et al. explores the molecular underpinnings of sexual dimorphism in antiviral immunity in zebrafish, with a particular emphasis on the male-biased gene cyp17a2. The authors demonstrate that male zebrafish exhibit stronger antiviral responses than females, and they identify a teleost-specific gene cyp17a2 as a key regulator of this dimorphism. Utilizing a combination of in vivo and in vitro methodologies, they demonstrate that Cyp17a2 potentiates IFN responses by stabilizing STING via K33-linked polyubiquitination and directly degrades the viral P protein via USP8-mediated

deubiquitination. The work challenges conventional views of sex-based immunity and proposes a novel, hormone- and sex chromosome-independent mechanism.

Strengths:

- (1) The following constitutes a novel concept, sexual dimorphism in immunity can be driven by an autosomal gene rather than sex chromosomes or hormones represents a significant advance in the field, offering a more comprehensive understanding of immune evolution.
- (2) The present study provides a comprehensive molecular pathway, from gene expression to protein-protein interactions and post-translational modifications, thereby establishing a link between Cyp17a2 and both host immune enhancement (via STING) and direct antiviral activity (via viral protein degradation).
- (3) In order to substantiate their claims, the authors utilize a wide range of techniques, including transcriptomics, Co-IP, ubiquitination assays, confocal microscopy, and knockout models.
- (4) The utilization of a singular model is imperative. Zebrafish, which are characterized by their absence of sex chromosomes, offer a clear genetic background for the dissection of autosomal contributions to sexual dimorphism.

Weaknesses:

- (1) Limited discussion on whether this mechanism extends beyond Cyprinidae and its implications for teleost adaptation.

Comments on revisions:

The authors successfully achieved their primary aim, which was to identify and characterize a male-biased gene governing antiviral sexual dimorphism in fish. The data provide robust support for the conclusion that Cyp17a2 enhances antiviral immunity through dual mechanisms, STING stabilization and viral protein degradation, independent of classical sex-determining pathways. The findings are consistent across a range of experimental setups and are statistically robust. The revisions have significantly enhanced the clarity, depth, and overall quality of the manuscript. The authors have addressed each concern meticulously, resulting in a much-improved and robust article. No further suggestions are offered.

<https://doi.org/10.7554/eLife.108048.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Weaknesses:

(1) Figure 10 outlines a mechanistic link between cyp17a2 and the sexual dimorphism the authors report for SVCV infection outcomes. The data presented on increased susceptibility of cyp17a2^{-/-} mutant male zebrafish support this diagram, but this conclusion is fairly weak without additional experimentation in both males and females. The authors justify their decision to focus on males by stating that they wanted to avoid potential androgen-mediated phenotypes in the cyp17a2 mutant background (lines 152156), but this appears to be speculation. It also doesn't preclude the possibility of testing the effects of increased cyp17a2 expression on viral infection in both males and females. This is of critical importance if the authors intend to focus the study on sexual dimorphism, which is how the introduction and discussion are currently structured.

Thank you for your suggestion. We have revised the relevant statements in the introduction and discussion sections accordingly. The cyp17a2 overexpression experiments were not conducted in both male and female individuals was primarily based on two reasons. First, our laboratory currently lacks the technical capability to achieve cyp17a2 overexpression at the organismal level, existing methodologies are limited to gene knockout via CRISPR-Cas9. Second, even if overexpression were feasible, subsequent comparisons would need to be restricted within sexes (i.e., female vs. female controls or male vs. male controls) to eliminate potential confounding effects of sex hormones. Such experimental outcomes would only demonstrate the antiviral function of Cyp17a2 itself rather than directly elucidate mechanisms underlying sexual dimorphism, which diverges from the central objective of this study.

We fully agree with your perspective and have accordingly refined relevant discussions in the revised manuscript. Our conclusions now emphasize that "cyp17a2 is one of the factors contributing to sex-based differences in antiviral immunity" rather than implying that it "solely mediates the entire phenotypic divergence." These modifications have been incorporated into the resubmitted version (Lines 112-115).

(2) The authors present data indicating an unexpected link between cyp17a2 and ubiquitination pathways. It is unclear how a CYP450 family member would carry out such activities, and this warrants much more attention. One brief paragraph in the discussion (starting at line 448) mentions previous implications of CYP450 proteins in antiviral immunity, but given that most of the data presented in the paper attempt to characterize cyp17a2 as a direct interactor of ubiquitination factors, more discussion in the text should be devoted to this topic. For example, are there any known domains in this protein that make sense in this context? Discussion of this interface is more relevant to the study than the general overview of sexual dimorphism that is currently highlighted in the discussion and throughout the text.

We are grateful to the reviewer for their suggestion to elaborate on this novel finding. The discussion on this point has been expanded significantly (Lines 448-460). It is acknowledged that Cyp17a2 is devoid of the canonical domains that are typically associated with the ubiquitination machinery (e.g., RING, U-box). The present study proposes that the endoplasmic reticulum (ER) localization of Cyp17a2, in conjunction with its capacity to function as a scaffold protein, is of paramount significance. By residing in the ER, Cyp17a2 is strategically positioned to interact with key immune regulators such as STING, which also

localizes to the ER. It is hypothesized that Cyp17a2 facilitates the recruitment of E3 ligases (btr32) and deubiquitinates (USP8) to their substrates (STING and SVCV P protein, respectively) by providing a platform for protein-protein interactions, rather than directly catalyzing ubiquitination. This noncanonical, scaffolding role for a cytochrome P450 (CYP450) enzyme represents an exciting evolutionary adaptation in teleost immunity.

(3) Figures 2-9 contain information that could be streamlined to highlight the main points the authors hope to make through a combination of editing, removal, and movement to supplemental materials. There is a consistent lack of clarity in these figures that could be improved by supplementing them with more text to accompany the supplemental figures. Using Figure 2 and an example, panel (A) could be removed as unnecessary, panel (B) could be exchanged for a volcano plot with examples highlighting why cyp17a2 was selected for further study and also the full dataset could be shared in a supplemental table, panel (C) could be modified to indicate why that particular subset was chosen for plotting along with an explanation of the scaling, panel (D) could be moved to supplemental because the point is redundant with panels (A) and (C), panel (E) could be presented as a heatmap, in panels (G) and (H) data from EPC cells could be moved to supplemental because it is not central to the phenotype under investigation, panels (J) to (L) and (N) to (P) could be moved to supplemental because they are redundant with the main points made in panels (M) and (Q). Similar considerations could be made with Figures 3-9.

We thank the reviewer for these excellent suggestions to improve the clarity and focus of our figures. A comprehensive review of all figures has been conducted in accordance with the recommendations made. Figure 2A has been removed. Figure 2B (revised Figure 2A) has been replaced with a volcano plot highlighting cyp17a2 and the full dataset has been provided as supplementary Table S2. Figure 2C (revised Figure 2B) is now a heatmap with eight sex-related genes and an explanation of the scaling has been added to the revised figure legends. Several panels (D, G, H, J-L, N-P) have been moved to the supplementary information (now Figure S1). Figure 2E has been presented as a heatmap. The same approach to streamlining has been applied to Figures 3-9, with confirmatory or secondary data being moved to supplements in order to better emphasize the main conclusions. The figure legends and main text have been updated accordingly.

(4) The data in Figure 3 (A)-(C) do not seem to match the description in the text. That is, the authors state that cyp17a2 overexpression increases interferon signaling activity in cells, but the figure shows higher increases in vector controls. Additionally, the data in panel (H) are not described. What genes were selected and why, and where are the data on the rest of the genes from this analysis? This should be shared in a supplemental table.

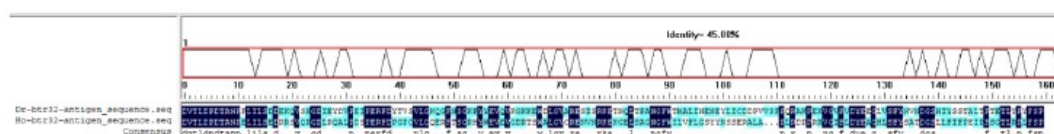
We apologize for the lack of clarity. In Figures 3A-C, the vector control shows baseline activation due to the stimulants (poly I:C/SVCV), but the fold-increase is significantly greater in the Cyp17a2-overexpressing groups. We have re-plotted the data to more clearly represent the stimulant-induced activation over baseline and added statistical comparisons between the Vector and Cyp17a2 groups under each condition to highlight the enhancing effect of Cyp17a2. For Figure 3H (revised Figure 3F), the heatmap shows a curated set of IFN-stimulated genes (ISGs) most significantly regulated by Cyp17a2 based on our RNA-seq analysis. We have added a description in the revised figure legend and in the results section (Lines 837-840). The full list of differentially expressed genes from this analysis is now provided in Supplementary Table S3.

(5) Some of the reagents described in the methods do not have cited support for the applications used in the study. For example, the antibody for TRIM11 (line 624, data in

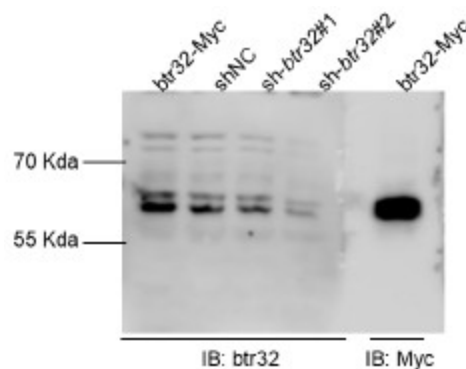
Figures 6 & 7) was generated for targeting the human protein. Validation for use of this reagent in zebrafish should be presented or cited. Furthermore, the accepted zebrafish nomenclature for this gene would be preferred throughout the text, which is *bloodthirsty-related gene family, member 32*.

We thank the reviewer for raising this important point regarding reagent specificity. To address the concern about antibody validation in zebrafish, we performed the following verification steps. First, we aligned the antigenic sequence targeted by the Abclonal btr32 antibody (ABclonal, A13887) with orthologous sequences from zebrafish, which showed 45% protein sequence similarity (Author response image 1). More importantly, we conducted experimental validation by expressing Myc-tagged btr32 in EPC cells. Both the anti-Myc and the anti-btr32 antibodies detected a protein band at the same molecular weight. Furthermore, when a btr32-specific knockdown plasmid was introduced, the band recognized by the anti-btr32 antibody was significantly reduced (Author response image 2). These results support the specificity of the antibody in recognizing fish btr32. In accordance with the reviewer's suggestion, we have also updated the gene nomenclature to "bloodthirsty-related gene family, member 32 (btr32)" throughout the manuscript.

Author response image 1.



Author response image 2.



Reviewer #2 (Public review):

Weaknesses:

(1) Colocalization analyses (Figures 4G, 6I, 9D) require quantitative metrics (e.g., Pearson's coefficients) rather than representative images alone.

We concur with the reviewer's assessment. We have now performed quantitative colocalization analysis (Pearson's coefficients) for all indicated figures (4G, 6I, 9D). The quantitative results are now presented within the figures themselves and described in the revised figure legends.

(2) Figure 1 survival curves need annotated statistical tests (e.g., "Log-rank test, $p=X.XX$ ")

The survival curves have now been annotated with the specific p-values from the Log-rank (Mantel-Cox) test (see revised Figures 1A, 2E).

(3) *Figure 2P GSEA should report exact FDR-adjusted p -values (not just " $p < 0.05$ ").*

Figure 2P (revised Figure S1J) has been updated to include the exact FDR p-values for the presented GSEA plots.

(4) *Section 2 overextends on teleost sex-determination diversity, condensing to emphasize relevance to immune dimorphism would strengthen narrative cohesion.*

The section on teleost sex-determination diversity in the Discussion (lines 357-365) has been condensed, with a more direct focus on how this diversity provides a unique context for studying immune dimorphism independent of canonical sex chromosomes, as exemplified by the zebrafish model.

(5) *Limited discussion on whether this mechanism extends beyond Cyprinidae and its implications for teleost adaptation.*

The discussion has been expanded (lines 375-386) to address the potential conservation of this mechanism. It is acknowledged that cyp17a2 is a teleost-specific gene, and it is hypothesized that its function in antiviral immunity may signify an adaptive innovation within this extensively diverse vertebrate group. It is suggested that further research in other teleost families will be essential to ascertain the broader evolutionary significance of the present findings.

Reviewer #2 (Recommendations for the authors):

(1) *Expand the Discussion to address why teleosts may have evolved male-biased immunity. Consider: pathogen pressure differentials in aquatic vs. terrestrial environments; trade-offs between immune investment and reproductive strategies (e.g., male-male competition); comparative advantages in external fertilization systems.*

We have expanded the discussion on lines 412-430, to address the potential conservation of this mechanism. We note that Cyp17a2 is a teleost-specific gene and speculate that its role in antiviral immunity represents an adaptive innovation within this highly diverse group of vertebrates. We propose that future studies of other teleost families are crucial for determining the broader evolutionary significance of our findings.

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